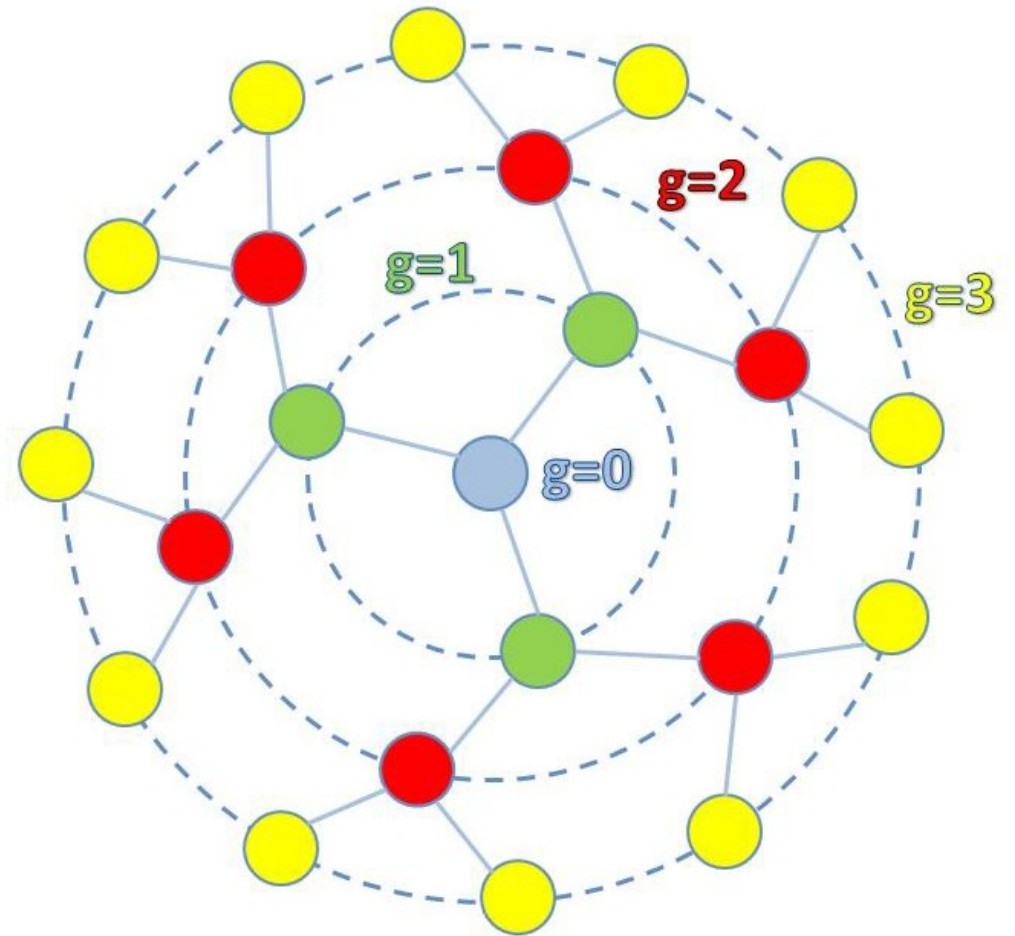
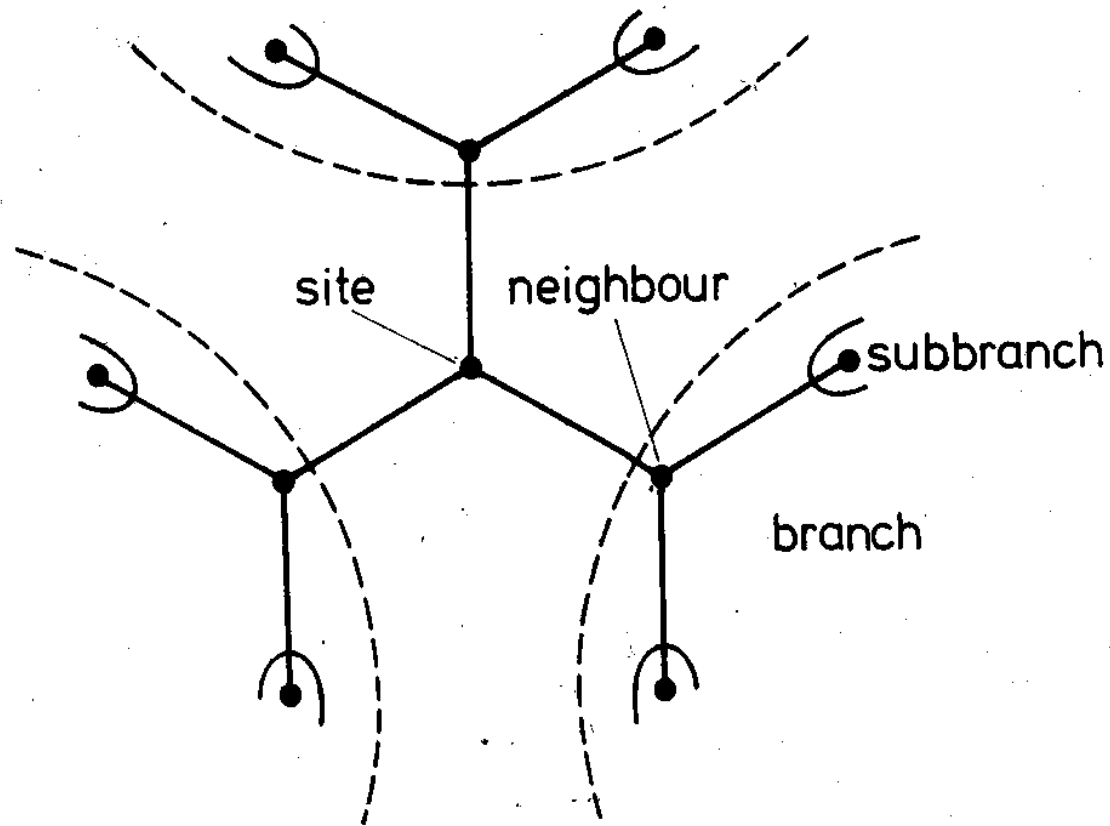


Historically, percolation theory goes back to Flory and to Stockmayer who during World War II used it to describe how small branching molecules form larger and larger macromolecules if more and more chemical bonds are formed between the original molecules. This polymerization process may lead to *gelation*, that is to the formation of a network of chemical bonds spanning the whole system. Thus the original small molecules correspond to our squares, the macromolecules to our clusters, and the network to our percolating cluster. You may be an experienced researcher in percolation without having been aware of it, for the boiling of an egg, which is first liquid and then becomes more solid-like ('gel') upon heating is an example. Flory and Stockmayer developed a theory which today one calls percolation theory on the Bethe lattice (or Cayley tree) and which will be explained later. But until recently it was controversial whether critical phenomena for gelation are described correctly by percolation theory and its assumption that chemical bonds are formed randomly (de Gennes, 1976; Kolb and Axelos, 1990).

Things you need to know to understand the percolation theory in any dimension

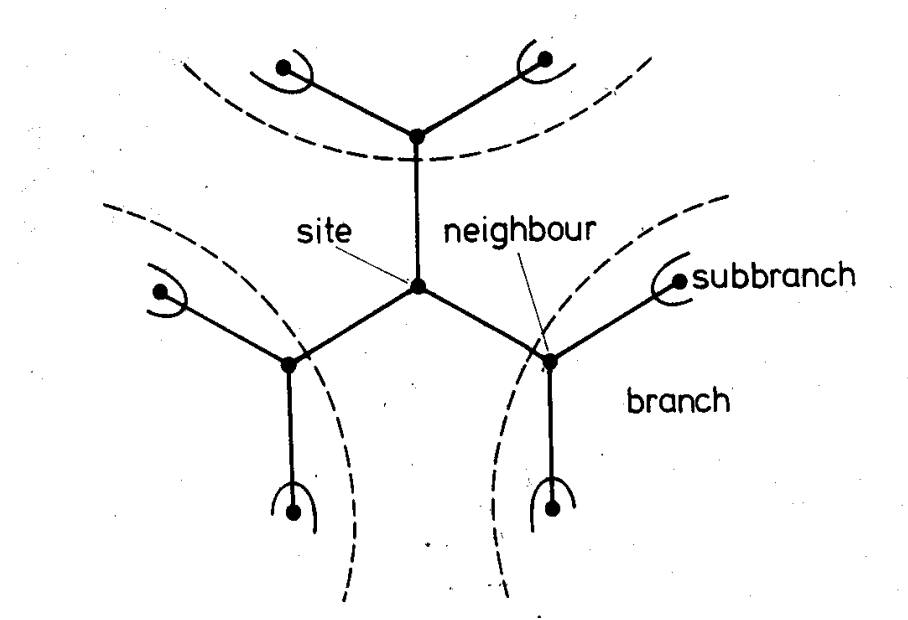
- Site and bond percolation
- Occupation probability
- $P(p; L)$ and $\Pi(p; L)$
- Cluster distribution
- Mean cluster size
- Exponents at critical point

Percolation in Bethe Lattice



Each site has z neighbouring sites
Each branch gives rise to $z - 1$ other branches
There are no closed loops
 $z=3$;

Percolation in Bethe Lattice



Q. Prove that, the number of surface sites relative to the total number of sites approaches a constant.

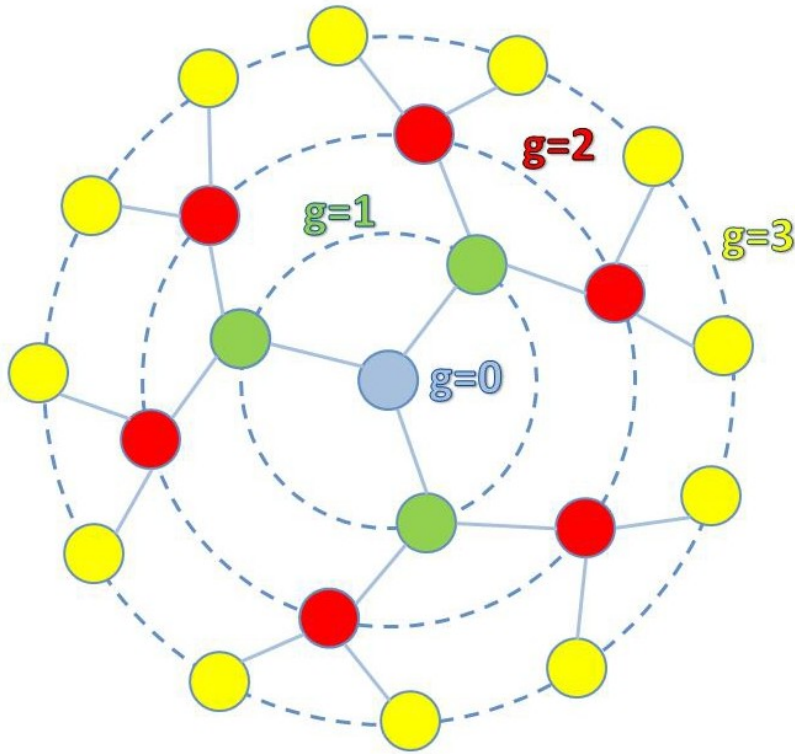
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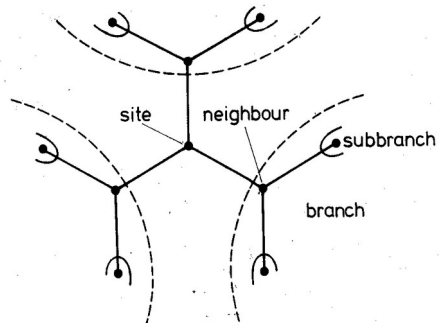
Percolation in Bethe Lattice



Q. Prove that, a) the number of surface sites relative to the total number of sites approaches a constant.

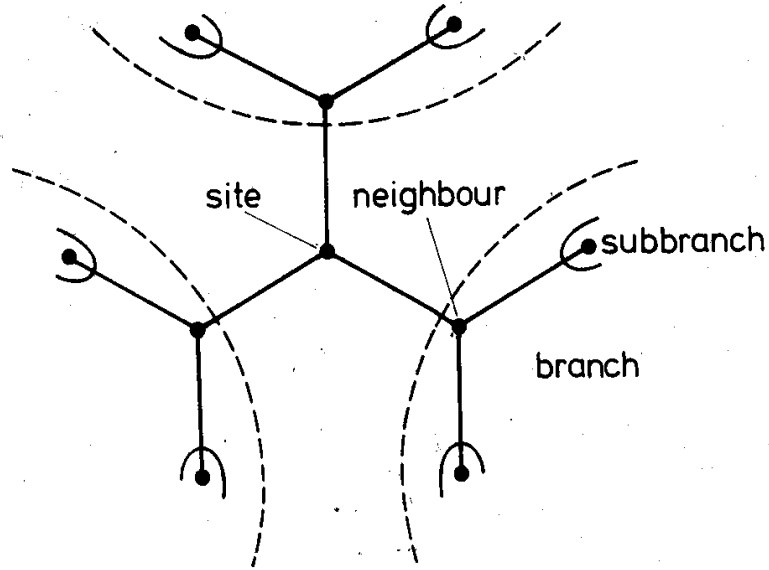
- Let g denote the **generation**, that is, the distance from a "centre site".
- Note, however, that in an infinite Bethe lattice, all sites are equivalent, so the notion of a "centre site" is not to be taken literally.
- In the figure left, the "first ring" of three sites belong to generation $g = 1$, the second "ring" of six sites belongs to the second generation and so on.

The total number of sites in a Bethe lattice consisting of g generations is



$$\begin{aligned}
 \text{Total no. sites} &= 1 + 3 \cdot (1 + 2 + \dots + 2^{g-1}) \\
 &= 1 + 3 \cdot \frac{1 - 2^g}{1 - 2} \\
 &= 3 \cdot 2^g - 2,
 \end{aligned}$$

Percolation in Bethe Lattice



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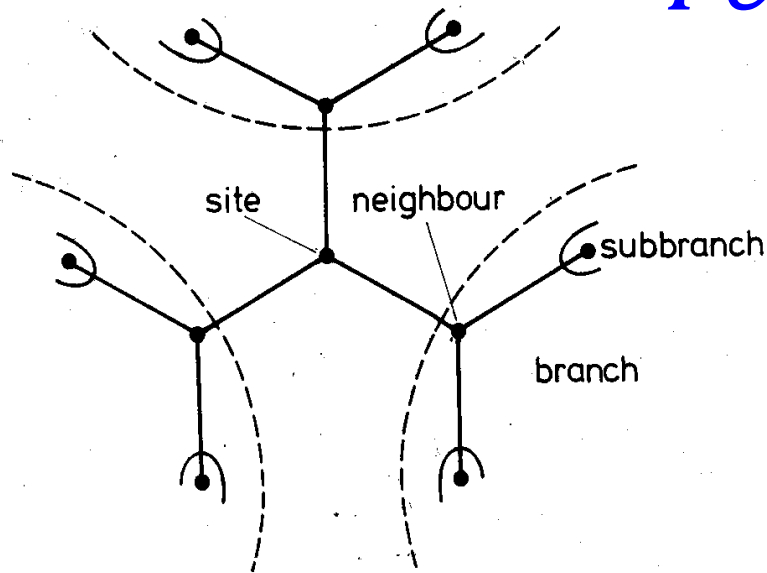
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 &= 3 \cdot 2^g - 2,
 \end{aligned}$$

while the number of surface sites is $3 \cdot 2^{g-1}$. Thus

$$\frac{\text{No. of surface sites}}{\text{Total no. of sites}} = \frac{3 \cdot 2^{g-1}}{3 \cdot 2^g - 2} \rightarrow \frac{1}{2} \quad \text{for } g \rightarrow \infty.$$

and the surface/volume tends to a constant.

Percolation in Bethe Lattice



What is the critical occupation probability in a Bethe lattice, that is, at which occupation probability does an infinite cluster (path) appear for the first time in an infinite Bethe lattice? Starting from a “centre site” and going outwards, we encounter $(z - 1)$ new neighbours. Thus, on average, we have $p(z - 1)$ *new* occupied sites on which we can continue the path. The critical occupation probability is determined by the equation

$$p_c(z - 1) = 1 \Leftrightarrow p_c = \frac{1}{z - 1}.$$

Notice, that for $z = 2$, the $1d$ percolation problem, we recover $p_c = 1$. For $z = 3$, e.g., $p_c = \frac{1}{2}$. Note that in a hypercubic lattice with $d \rightarrow \infty$ one would expect $p_c \rightarrow 1/(2d - 1)$ in order to be able to continue walking along a path of occupied sites.

Percolation in Bethe Lattice

The strength of the infinite cluster $P(p)$ is the probability that an arbitrary site belongs to the infinite cluster

Let define the quantity Q as the probability that an arbitrary site is NOT connected to infinity through a fixed branch originating at this site.

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$$P(p) = (\text{Prob. site is occupied}) \times (\text{Prob. at least ONE branch lead to infinity}) = p \times (1 - Q^3)$$

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It is quadratic, and the solution of Q will be

$$Q = \frac{1 \pm \sqrt{(2p - 1)^2}}{2p} = \begin{cases} 1 \\ \frac{1-p}{p} \end{cases}$$

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Using a Taylor expansion for $P(p)$ around $p = p_c = \frac{1}{2}$, it can be shown that

$$P(p) \propto (p - p_c) \quad \text{for } p \rightarrow p_c^+.$$

$$\frac{dP_\infty}{dp} = \left[-3 \left(\frac{1-p}{p} \right)^2 \frac{-p-1+p}{p^2} \right] p$$

$$P(p) \approx \left[p \times 3 \left(\frac{1-p}{p} \right)^2 \frac{1}{p^2} + 0 \right]_{p=p_c} (p - p_c) = 6(p - p_c)$$

Percolation in Bethe Lattice

Mean Cluster size

What is the mean cluster size $S(p)$ to which an occupied site (e.g. the “centre site”) belongs?

Let T be the mean cluster size in one branch.

$$\begin{aligned}\text{Thus, } S(p) &= \text{average cluster size to which the origin belongs} \\ &= (\text{contribution from origin}) + (\text{contributions from the 3 branches}) \\ &= 1 + 3T\end{aligned}$$

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If the neighbour to the “centre site” is empty, probability $(1 - p)$ there is no contribution to T from this branch. If the neighbour to the “centre site” is occupied (probability p), it contributes its own mass (unity) to the cluster, and adds the mass T for each of its $z - 1 = 2$ subbranches. Thus,

$$T = (1 - p) \cdot 0 + p \cdot (1 + 2T) \Leftrightarrow T = \frac{p}{1 - 2p} \quad \text{for } p < p_c.$$

Percolation in Bethe Lattice

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$$S(p) = 1 + 3T = 1 + \frac{3p}{1 - 2p} = \frac{1 + p}{1 - 2p}$$

$$S(p) = 1 + 3T = 1 + \frac{3p}{1 - 2p} = \frac{1 + p}{2 \times \frac{1}{2} - 2p} = \frac{1 + p}{2} (p_c - p)^{-1}$$

mean cluster of size $S(p)$ diverges for $p \rightarrow p_c$

Critical phenomena also occur for thermal phase transitions; the Bethe lattice approximation for percolation theory then is somewhat analogous to the molecular field approximation for magnetism, or the van der Waals equation for fluids. (More precisely, one should use the better Bethe approximation for magnetism, which is exact on the Bethe lattice but has the same critical exponents as the molecular field approximation.) In all three cases, rather simple formulae for the order parameter can be derived exactly, leading to a simple power law very near to the critical point. For example, the difference between liquid and vapour density, or the spontaneous magnetization, vanish in both approximations as $(T_c - T)^{1/2}$. Thus the analogy between thermal critical phenomena and percolation is not complete since the critical exponent for the order parameter is $1/2$ for thermal phase transitions and unity for percolation, in these approximations.

Percolation in 2D lattice

Percolation in 2-d

- ❖ Percolation theory is a branch of statistical physics that deals with the behavior of connected clusters in random systems. The basic idea is to study **how connectivity emerges in a disordered medium**. The model involves a **lattice or network of sites**, and each site is **occupied** with a certain probability (p) or **vacant** with the complementary probability ($1 - p$). p is called as **occupation probability**.

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- ❖ The most common example is the **percolation on a square lattice**, where sites are connected to their nearest neighbors. ***If a cluster of connected sites spans the lattice from one side to another, it is said to percolate.*** The percolation phase transition occurs at a **critical probability (p_c)**, separating a regime where isolated clusters dominate from a regime where a **giant percolating cluster emerges**.

Percolation in 2-d

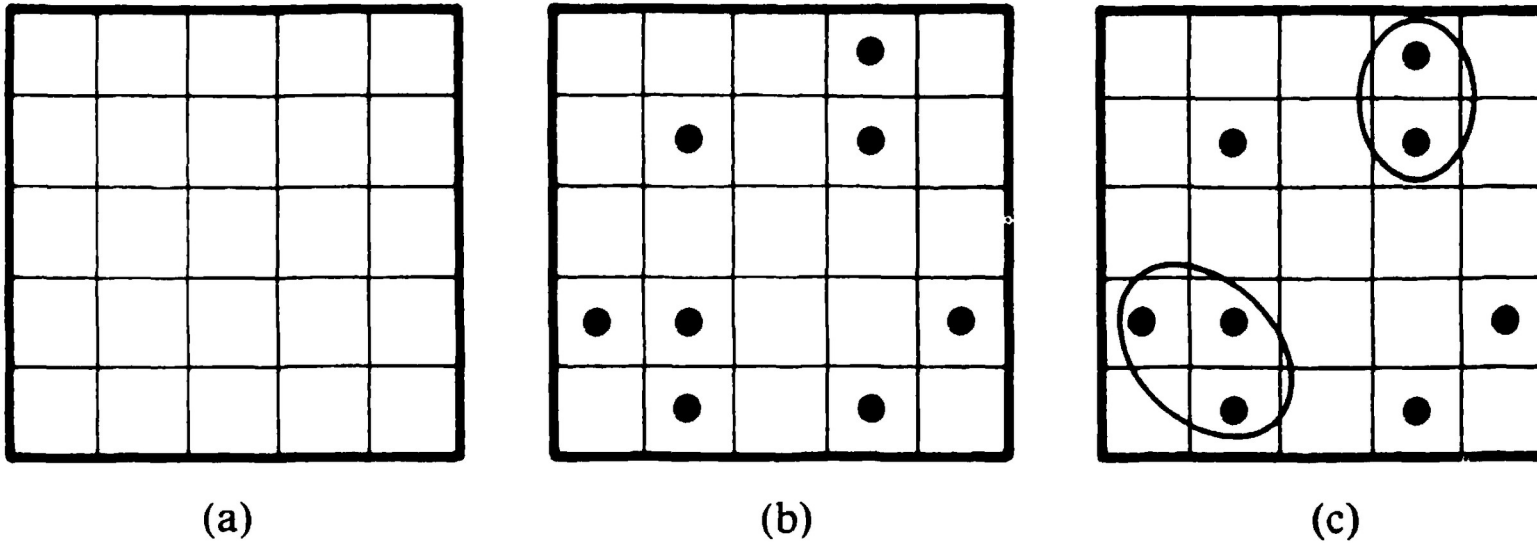
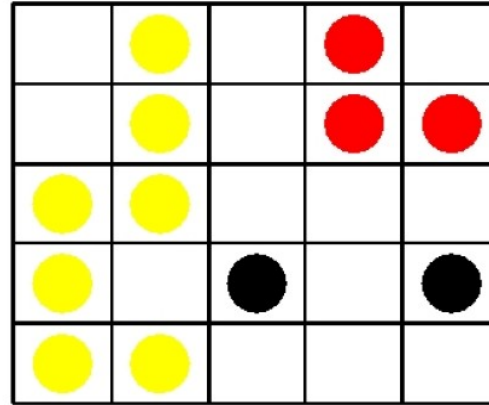


Fig. 1. Definition of percolation and its clusters: (a) shows parts of a square lattice; in (b) some squares are occupied with big dots; in (c) the 'clusters', groups of neighbouring occupied squares, are encircled except when the 'cluster' consists of a single square.

Percolation in 2-d



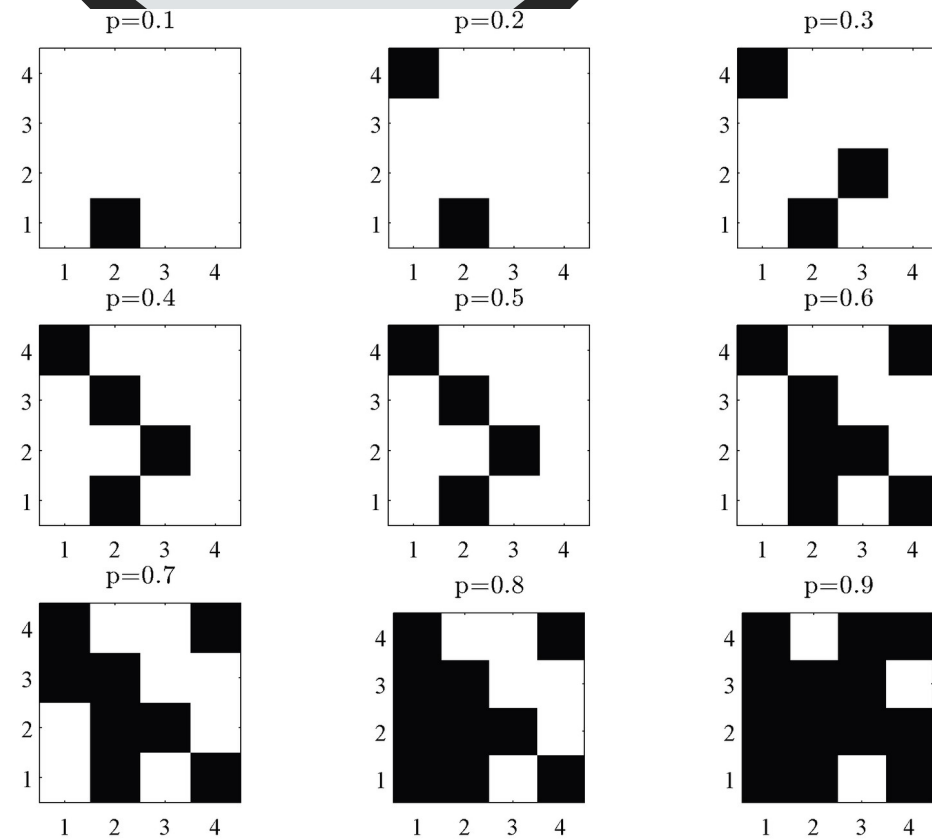
Percolation in $2d$ square lattice of linear size $L = 5$. Sites are occupied with probability p . In the lattice above, we have one cluster of size 7, a cluster of size 3 and two clusters of size 1 (isolated sites).

A cluster is a group of nearest neighbouring occupied sites.

Percolation in 2-d

The figure suggests p_c depends on the details:

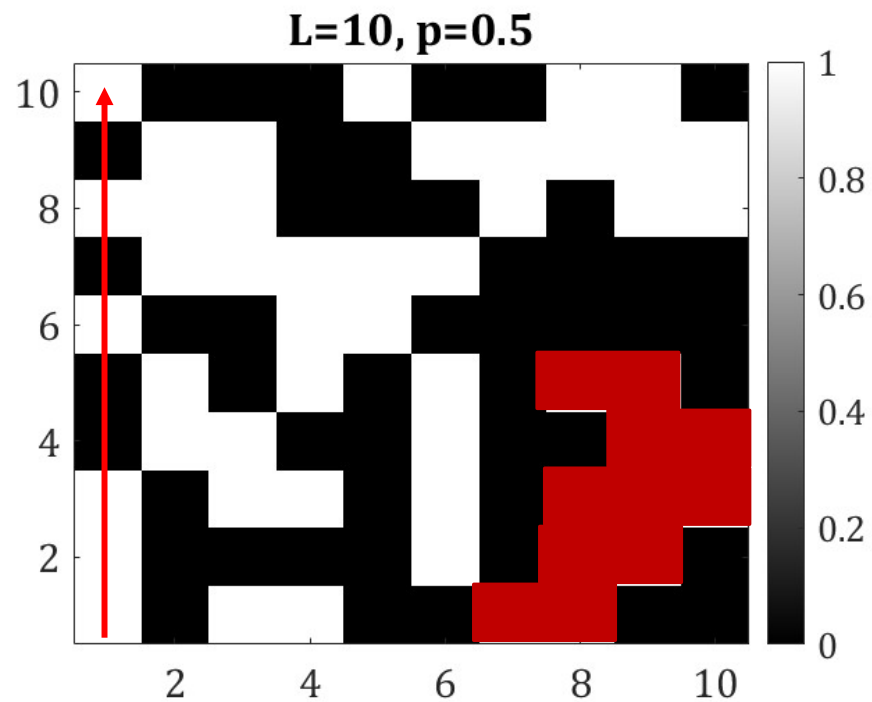
- It depends on the rule for connectivity.
- It would also depend on the type of lattice used: square, triangular, hexagonal, etc.
- The value of p_c is specific.



However, many other properties are general. For example, how the conductivity of the porous medium depends on p . Near p_c does not depend on the type of lattice or the choice of connectivity rule. **It is universal**

Percolation in 2-d

```
L = 20;  
p = 0.5;  
z = rand(L,L);  
m = z < p;  
imagesc(m);
```

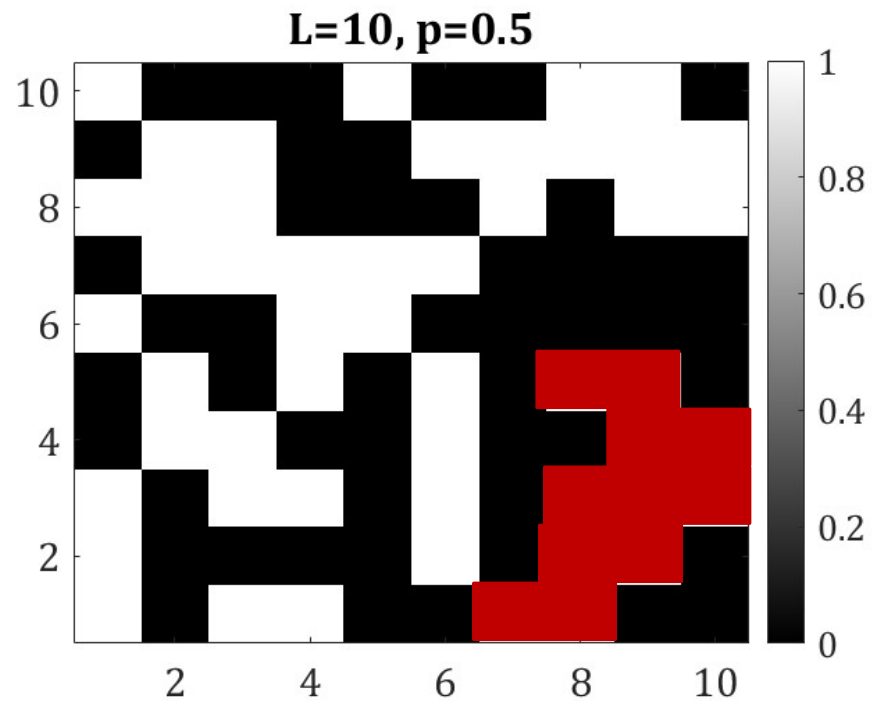


Marking the clusters

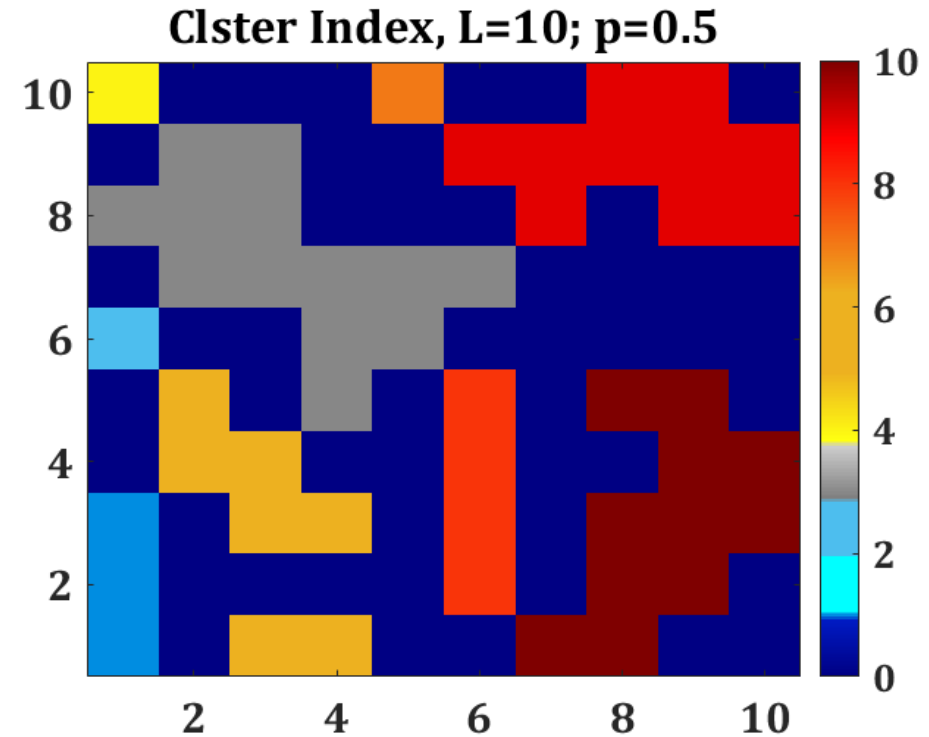
	1	2	3	4	5	6	7	8	9	10
1	1	0	6	6	0	0	10	10	0	0
2	1	0	0	0	0	8	0	10	10	0
3	1	0	5	5	0	8	0	10	10	10
4	0	5	5	0	0	8	0	0	10	10
5	0	5	0	3	0	8	0	10	10	0
6	2	0	0	3	3	0	0	0	0	0
7	0	3	3	3	3	3	0	0	0	0
8	3	3	3	0	0	0	9	0	9	9
9	0	3	3	0	0	9	9	9	9	9
10	4	0	0	0	7	0	0	9	9	0

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```

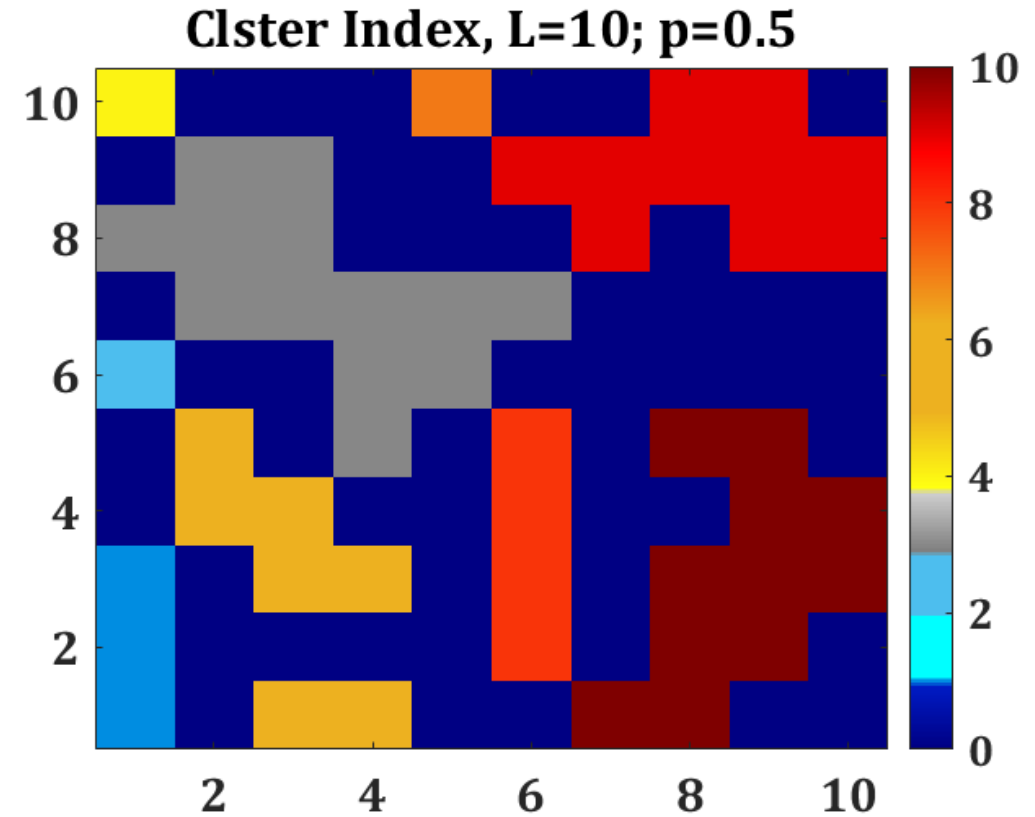


```
[lw,num] = bwlabel(m,4);  
imagesc(lw)
```



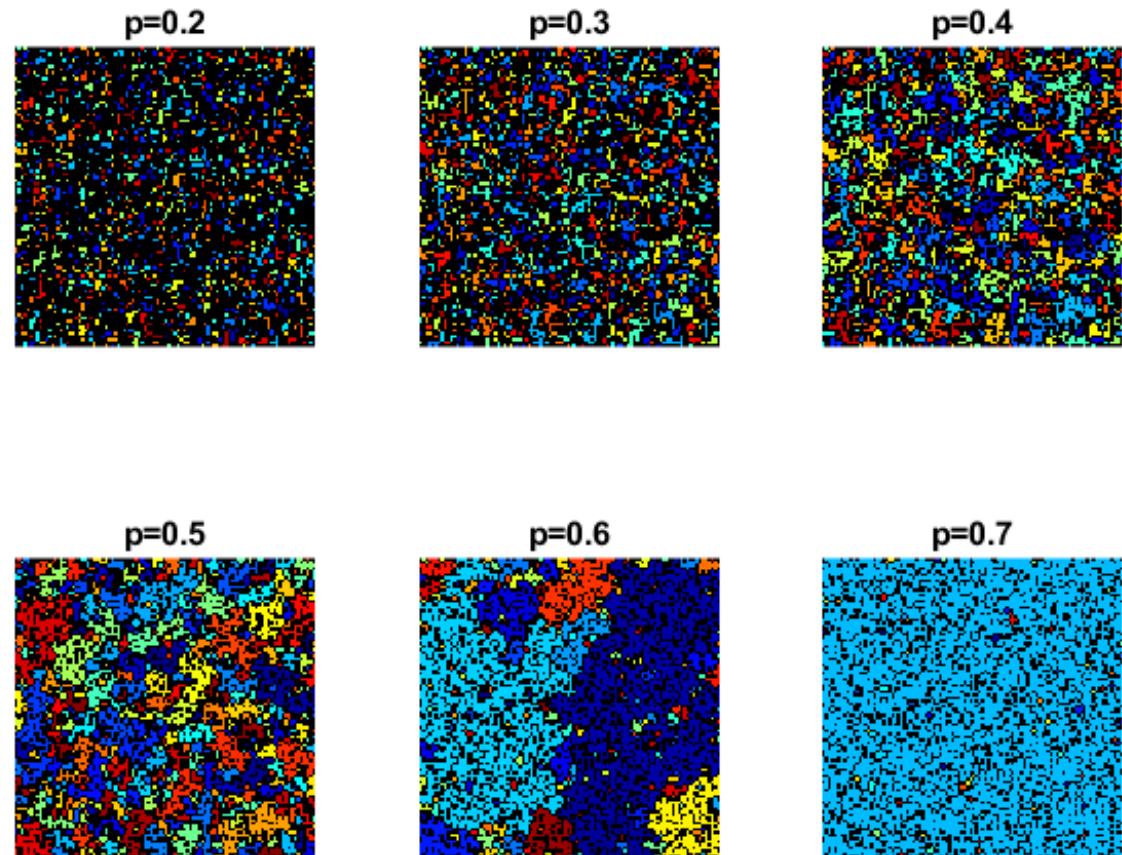
Percolation in 2-d

	1	2	3	4	5	6	7	8	9	10
1	1	0	6	6	0	0	10	10	0	0
2	1	0	0	0	0	8	0	10	10	0
3	1	0	5	5	0	8	0	10	10	10
4	0	5	5	0	0	8	0	0	10	10
5	0	5	0	3	0	8	0	10	10	0
6	2	0	0	3	3	0	0	0	0	0
7	0	3	3	3	3	3	0	0	0	0
8	3	3	3	0	0	0	9	0	9	9
9	0	3	3	0	0	9	9	9	9	9
10	4	0	0	0	7	0	0	9	9	0



Each of the color represents each disjoint cluster

*Percolation in 2-d
(Numerical
methods:
Percolating clusters)*



Each of the color represents each disjoint cluster

Percolation Theory

Two sites are connected if they are nearest neighbors (4 neighbors on square lattice)

- *a cluster is a set of connected sites*
- *a cluster is **spanning** if it spans from one side to the opposite side*
- *a cluster that is spanning is called the **spanning cluster***
- *a system is **percolating** if there is a spanning cluster in the system*

Someone who is studying percolation process will focus on

- *characterization of the sizes of connected regions,*
- *the size of the region connecting one side to another,*
- *the size of the region that contributes to transport (fluid, thermal or electrical transport), and other geometrical properties of the system.*

Most of the features we study will be universal – independent of many of the details of the system.

- ❖ Percolation theory is a branch of statistical physics that deals with the behavior of connected clusters in random systems. The basic idea is to study **how connectivity emerges in a disordered medium**. The model involves a **lattice or network of sites**, and each site is **occupied** with a certain probability (p) or **vacant** with the complementary probability ($1 - p$). p is called as **occupation probability**.
- ❖ The most common example is the **percolation on a square lattice**, where sites are connected to their nearest neighbors. ***If a cluster of connected sites spans the lattice from one side to another, it is said to percolate.*** The percolation phase transition occurs at a **critical probability (p_c)**, separating a regime where isolated clusters dominate from a regime where a **giant percolating cluster emerges**.
- ❖ *Understanding percolation theory involves **studying the critical behavior near the percolation threshold**, where **small** changes in the probability of occupation can lead to **drastic** changes in the system's macroscopic properties. This behavior is often characterized by **critical exponents and scaling laws**.*

Percolation in 2-d

- ❖ *Fractals, which are objects with **self-similar structure** at different scales, are intimately related to percolation theory. **The clusters formed during percolation often exhibit fractal characteristics**, and the study of percolation provides insights into the formation of fractal structures in various physical systems.*
- ❖ *Scaling is a key concept in percolation theory, describing how macroscopic properties of a system depend on its size. **Near the critical point, certain properties exhibit scaling behavior**, where they can be expressed as power laws. The renormalization group theory is a powerful tool used to understand the scaling behavior and critical phenomena in percolation and other systems.*
- ❖ *The **application** of percolation theory extends beyond physics to various fields, including biology, geophysics, and even practical applications like oil recovery. In biology, percolation models have been used to study processes like disease spreading in populations. In geophysics, percolation is relevant for understanding phenomena like fluid flow through porous rock. In oil recovery, percolation concepts can be applied to optimize the extraction of oil from reservoirs.*

Percolation in 2-d

A cluster is a group of nearest neighbouring occupied sites.

The cluster number $n_s(p)$ denotes the number of s-clusters per lattice site.

The percolation threshold p_c is the concentration (occupation probability) p at which an infinite cluster appears for the first time in an infinite lattice.

Caveats

- In the one-dimensional case it was simple to find $\Pi(p, L)$ because there is only one possible path from one side to another. We cannot generalize this to two dimensions, since in two-dimensions there are many paths from one side to another – and we need to include all to estimate the probability for percolation.
- Similarly, it was simple to find $n(s, p)$, because all clusters only have two neighboring sites – the surface is always of size 2.
- This is also not generalizable to higher dimensions.

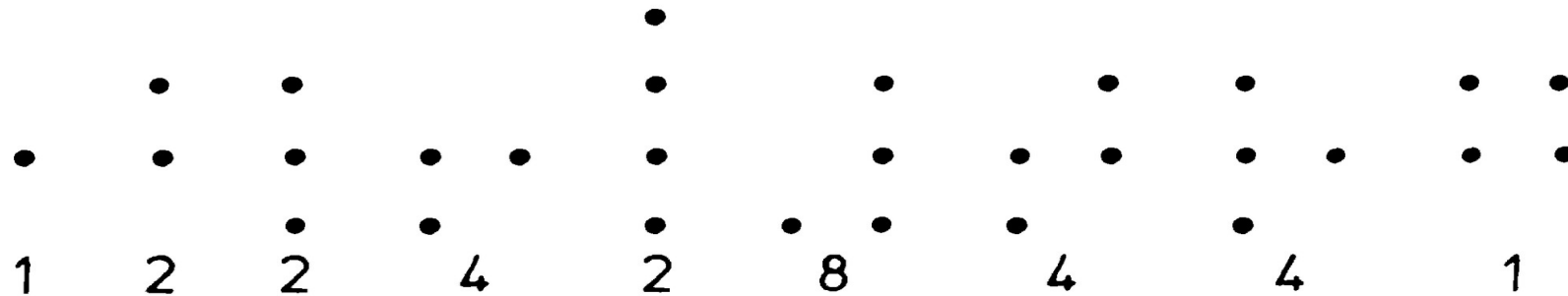
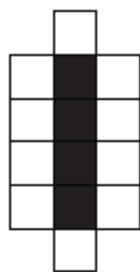


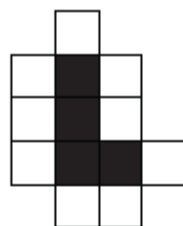
Fig. 8. List of all cluster configuration ('lattice animals') on the square lattice up to $s = 4$. For each structure, mirror images and rotated configurations are not shown; only the total number of such configurations, including the one shown, is indicated under each structure.



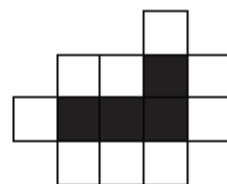
$s=4, t=10$



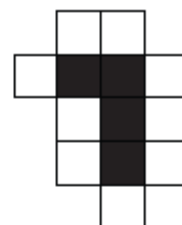
$s=4, t=10$



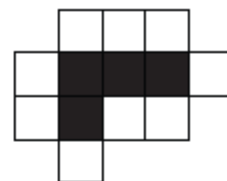
$s=4, t=9$



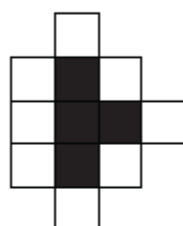
$s=4, t=9$



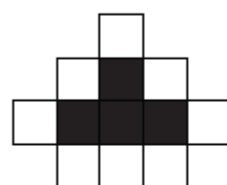
$s=4, t=9$



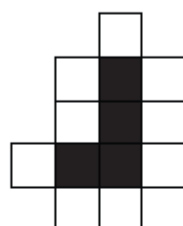
$s=4, t=9$



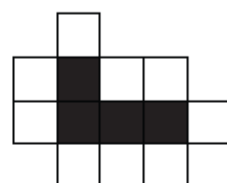
$s=4, t=8$



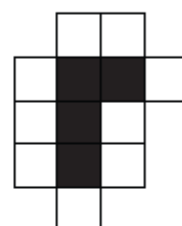
$s=4, t=8$



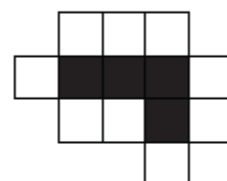
$s=4, t=9$



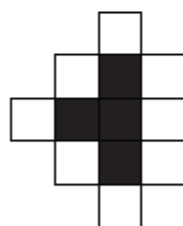
$s=4, t=9$



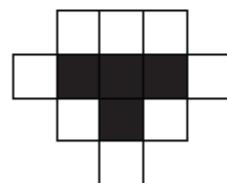
$s=4, t=9$



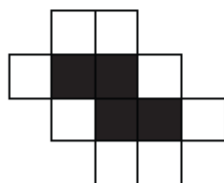
$s=4, t=9$



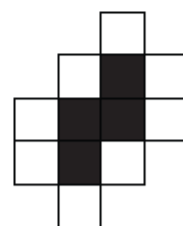
$s=4, t=8$



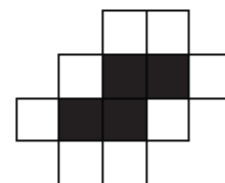
$s=4, t=8$



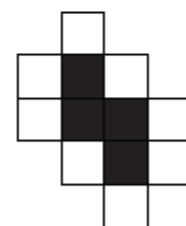
$s=4, t=8$



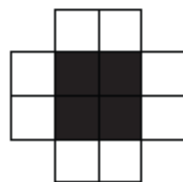
$s=4, t=8$



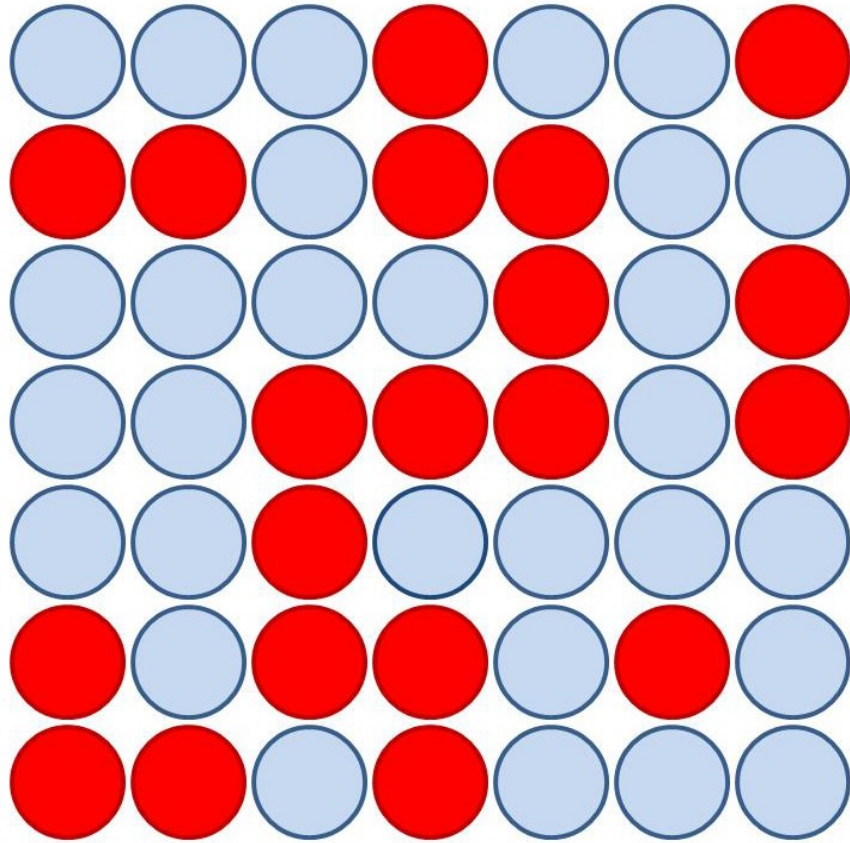
$s=4, t=8$



$s=4, t=8$



$s=4, t=8$



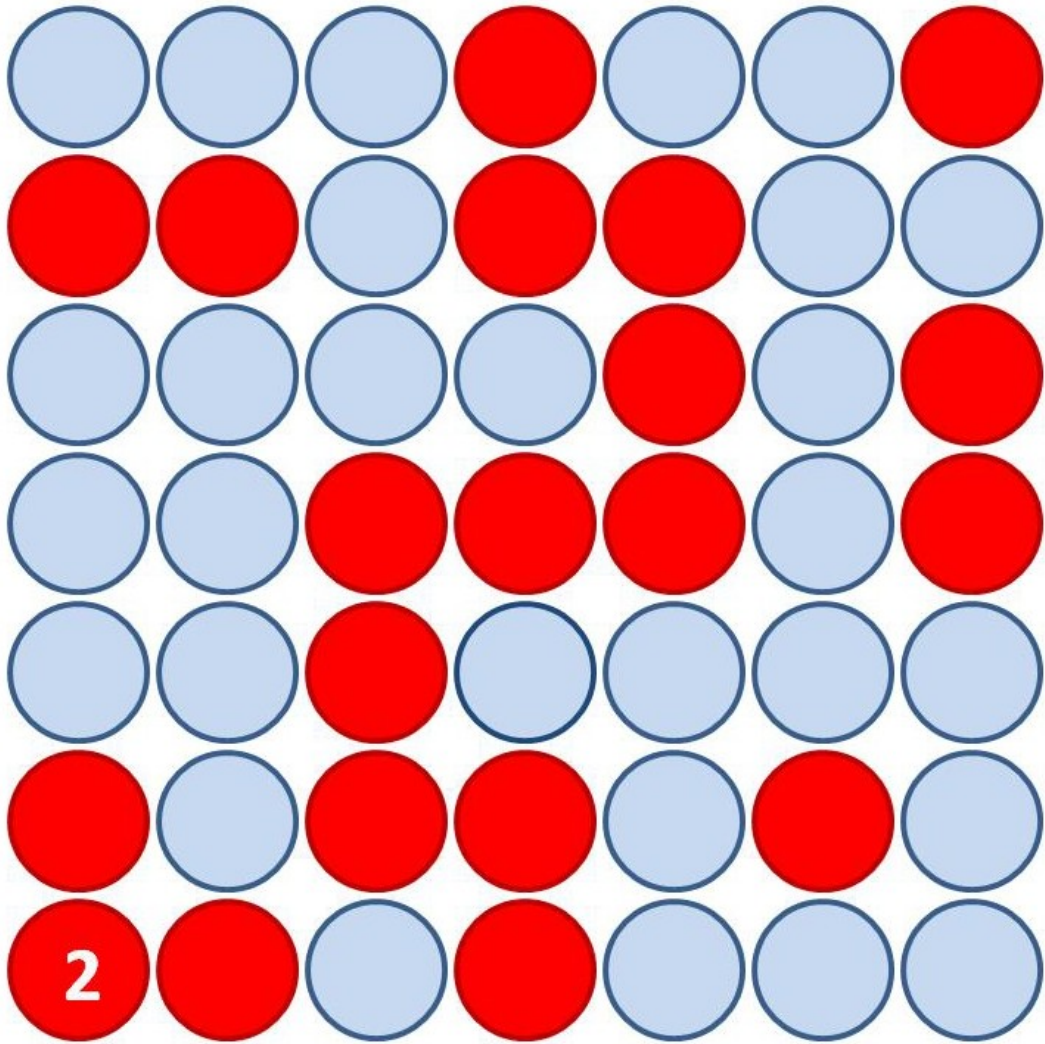
1. **start** from the site in the **left-bottom corner**;
2. **sweep** from **left to right bottom to top**;
3. **only** verify **left** and **bottom** neighbors.

$$k = 2$$

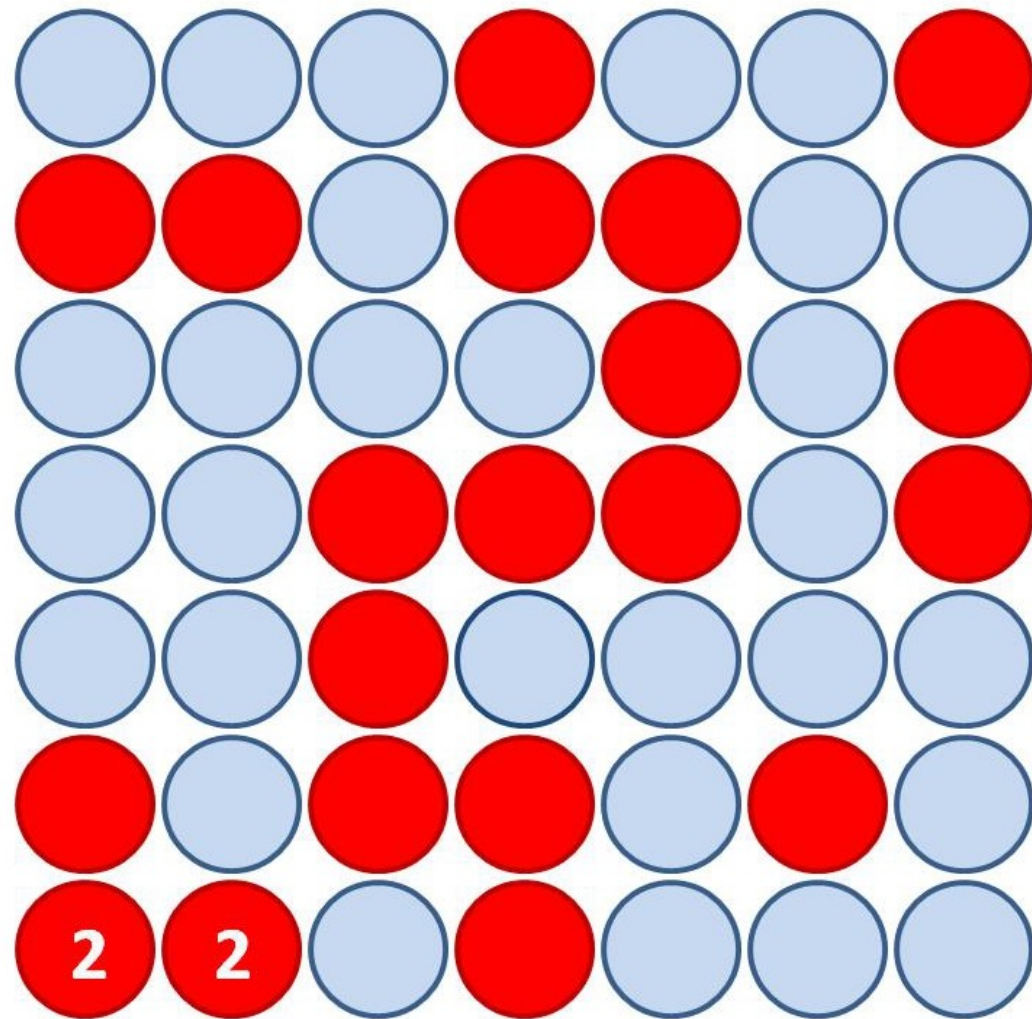
$$M(k) = 0$$

Nuno Araújo

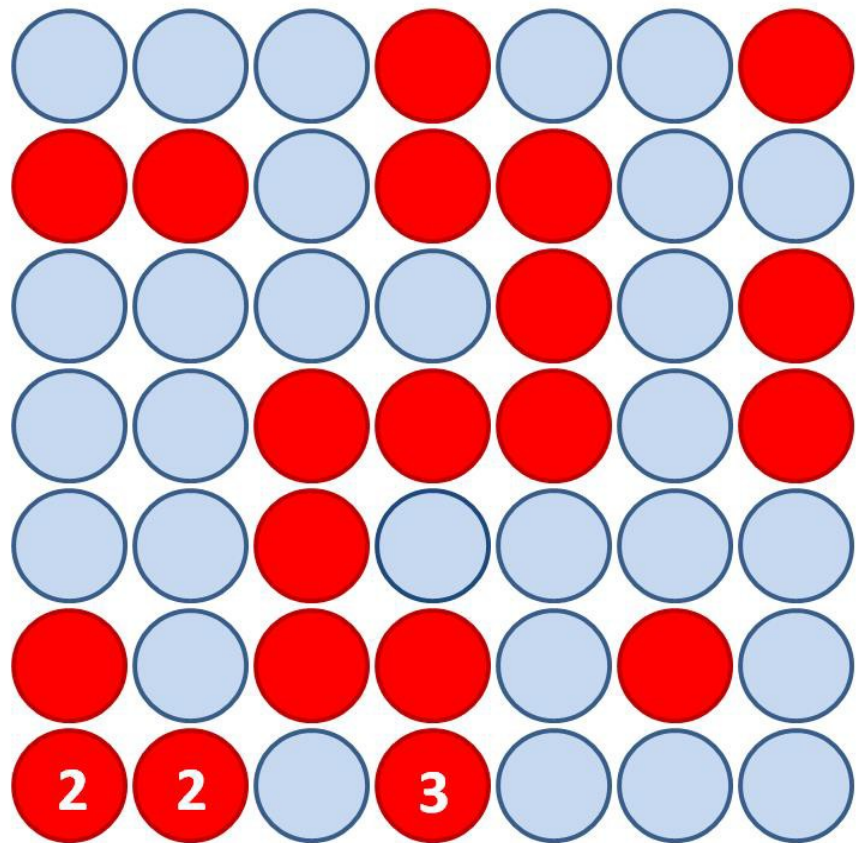
J. Hoshen and R. Kopelman. *Phys. Rev. B* **14**, 3438 (1976)



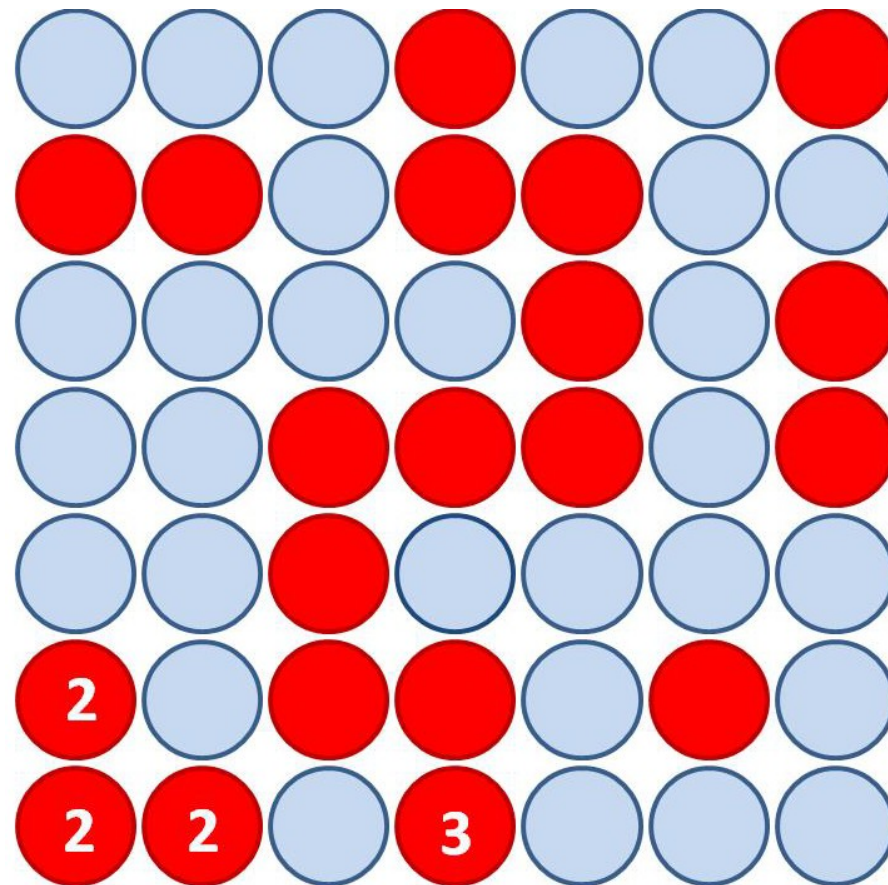
k	M(k)
2	1



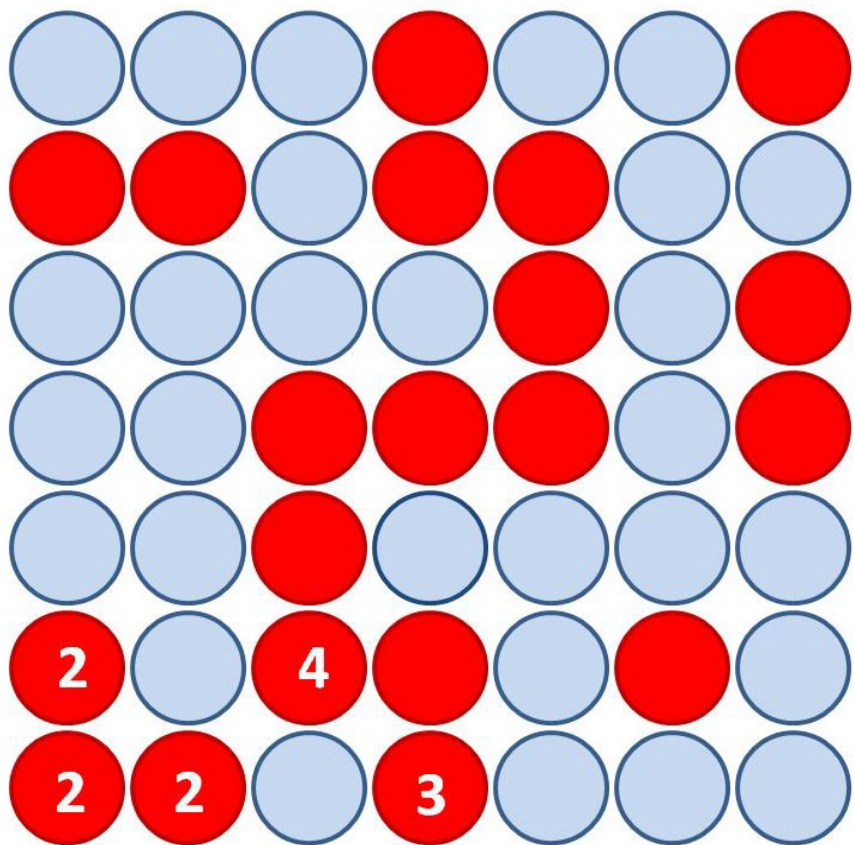
k	M(k)
2	2



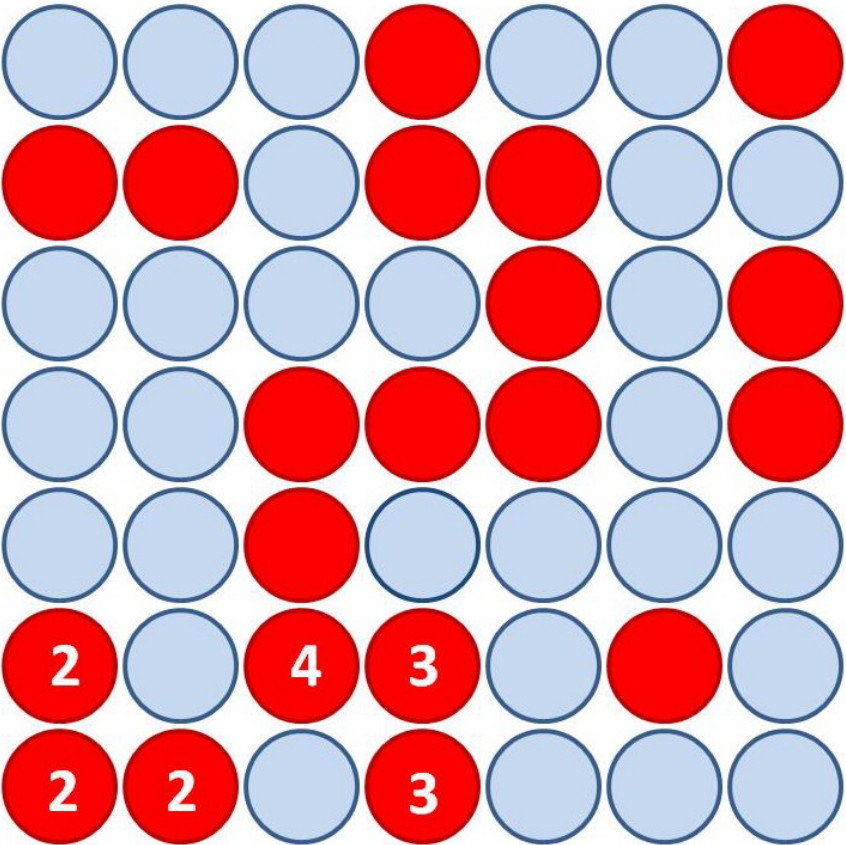
k	M(k)
2	2
3	1



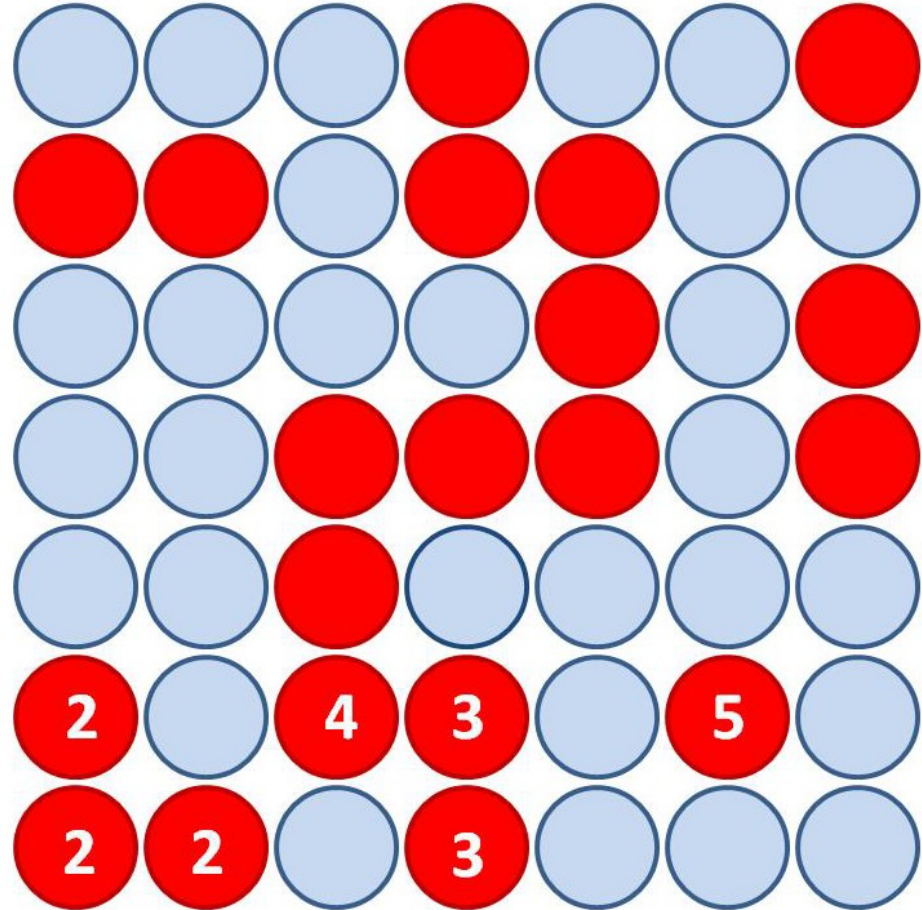
k	M(k)
2	3
3	1



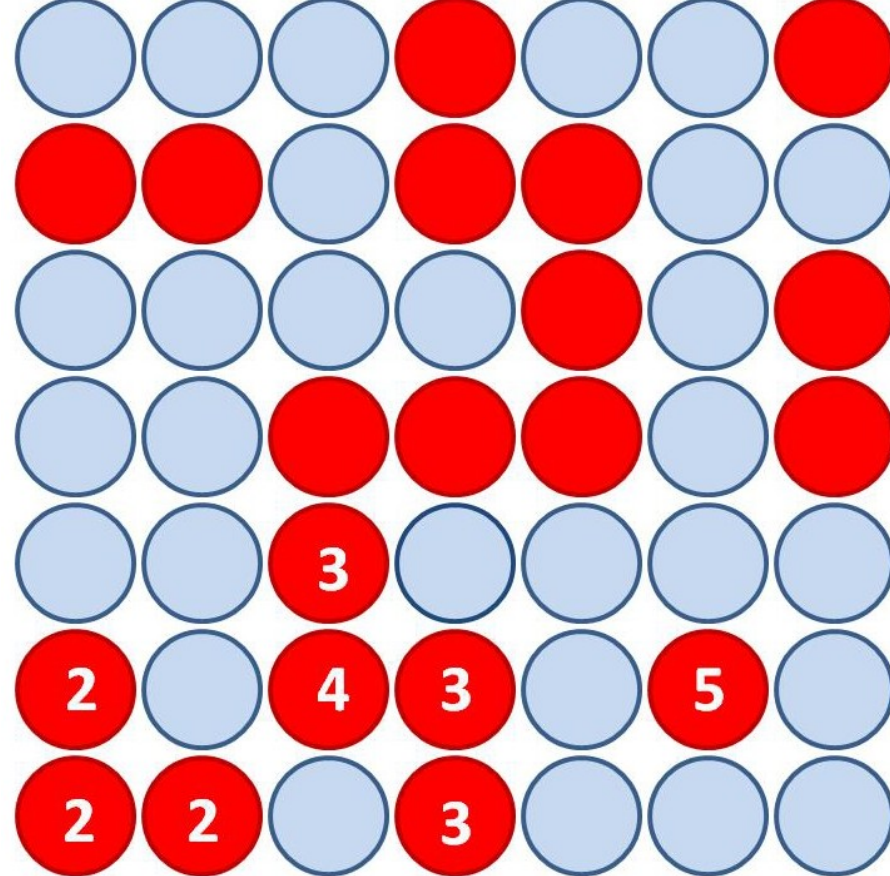
k	M(k)
2	3
3	1
4	1



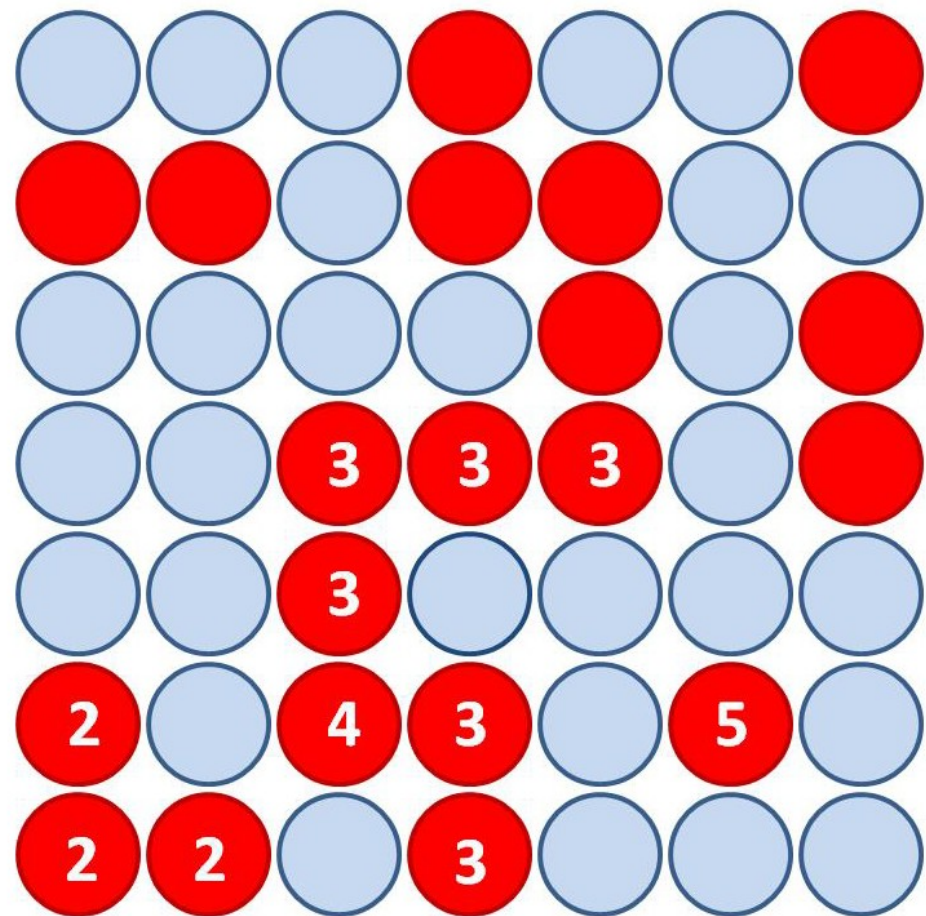
k	M(k)
2	3
3	3
4	-3



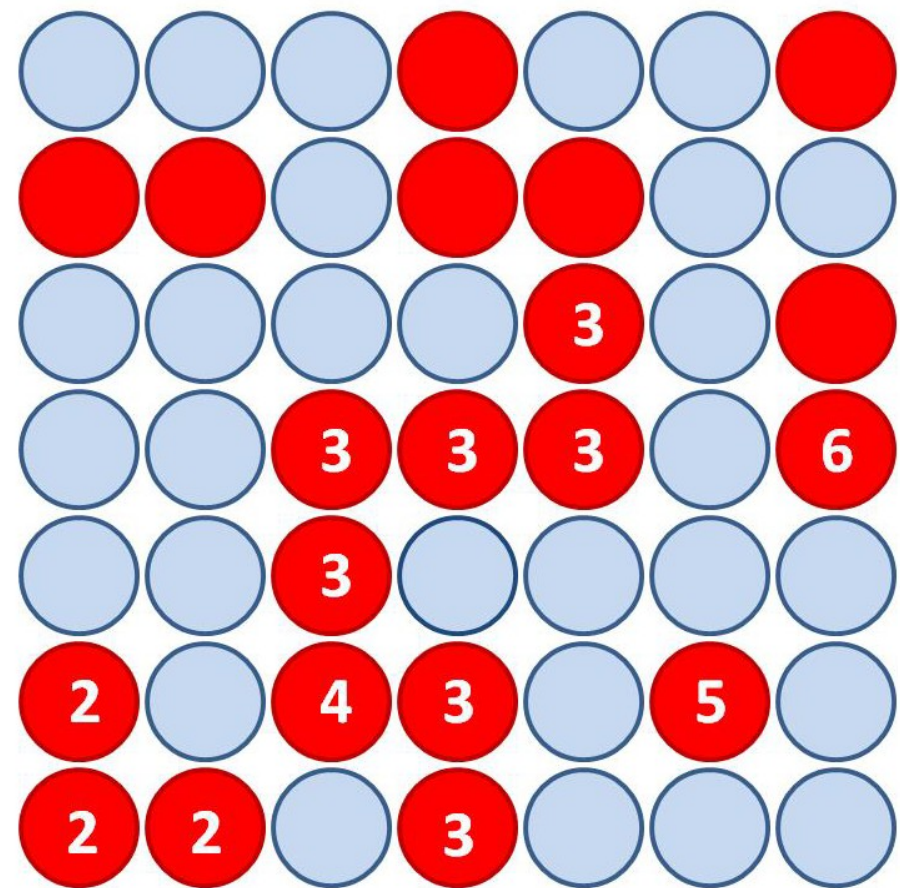
k	M(k)
2	3
3	3
4	-3
5	1



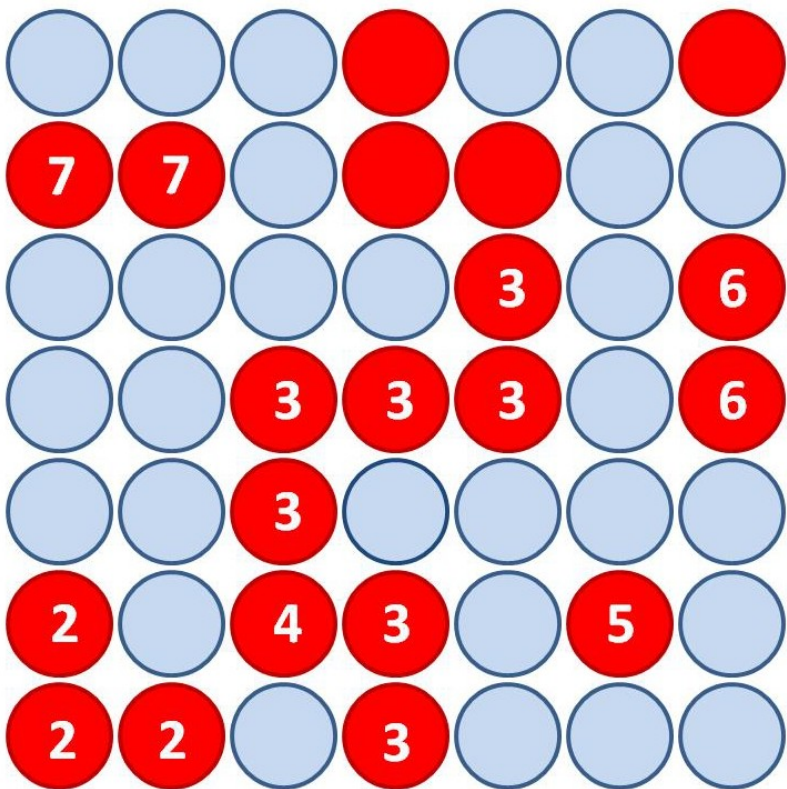
k	M(k)
2	3
3	4
4	-3
5	1



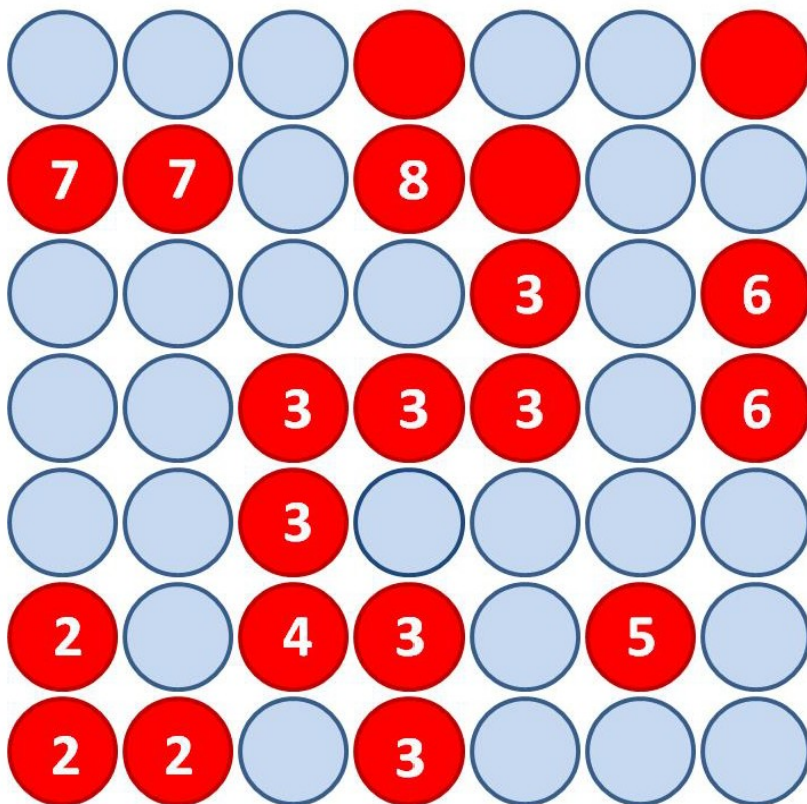
k	M(k)
2	3
3	7
4	-3
5	1



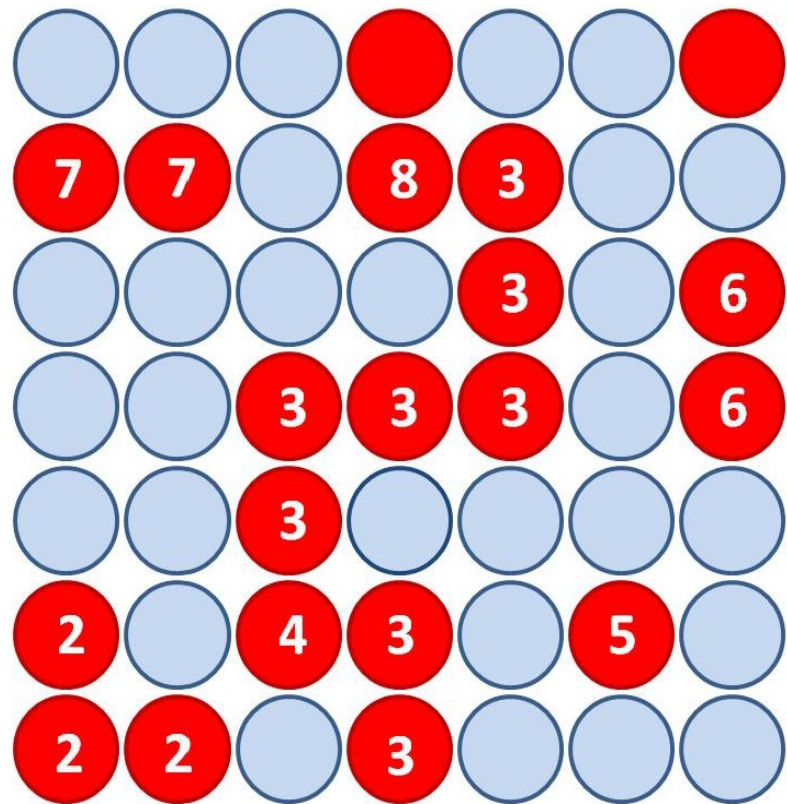
k	M(k)
2	3
3	8
4	-3
5	1
6	1



k	M(k)
2	3
3	8
4	-3
5	1
6	2
7	2



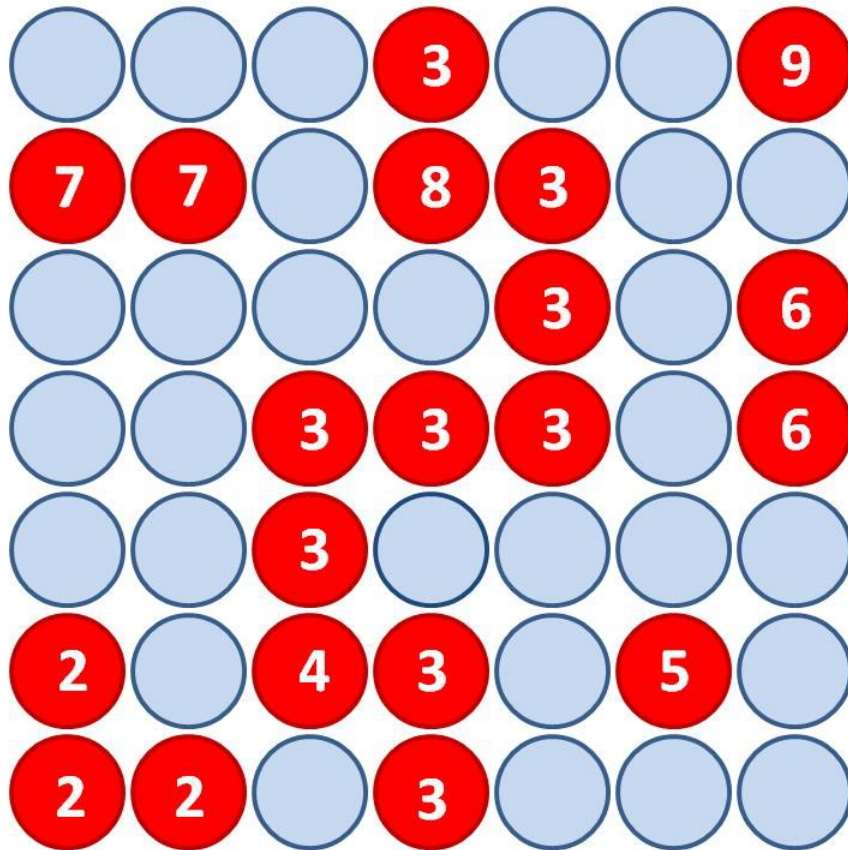
k	M(k)
2	3
3	8
4	-3
5	1
6	2
7	2
8	1



k	M(k)
2	3
3	10
4	-3
5	1
6	2
7	2
8	-3

Algorithms

Hoshen and Kopelman

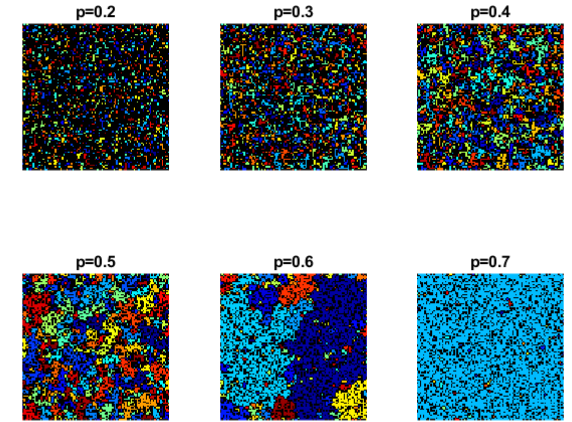


1. **start** from the site in the **left-bottom corner**;
2. **sweep** from **left to right**
bottom to top;
3. **only** verify **left** and **bottom** neighbors.

k	M(k)
2	3
3	11
4	-3
5	1
6	2
7	2
8	-3
9	1

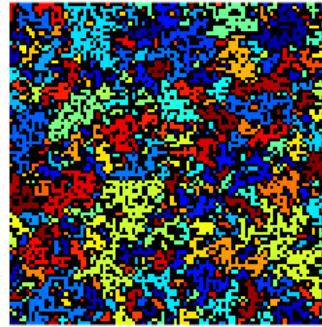
*Percolation in 2-d
(Numerical
methods:
Percolating clusters)*

```
L = 100;
pv = [0.2 0.3 0.4 0.5 0.6 0.7];
z = rand(L,L);
for i = 1:length(pv)
    p = pv(i);
    m = z < p;
    [lw,num] = bwlabel(m,4); %%% marking the clusters
    mat = lw;
    img = label2rgb(lw,'jet','k','shuffle');
    subplot(2,3,i)
    tit=sprintf('p=%0.5g',p);
    image(img); axis square;
    title(tit);
    axis off
end
```

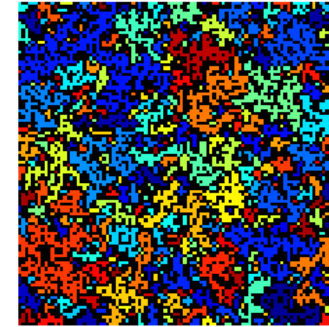


*Percolation in 2-d
(Percolating
clusters) Near to
critical points*

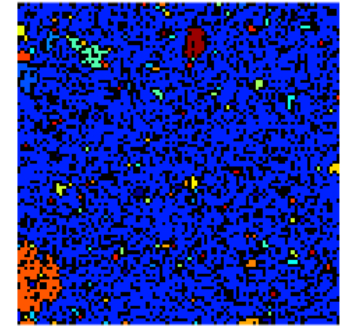
$p=0.5$



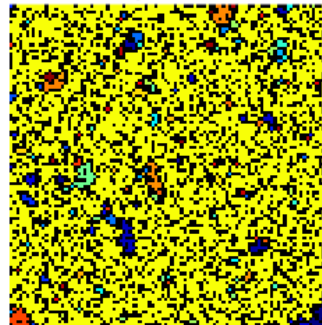
$p=0.5$



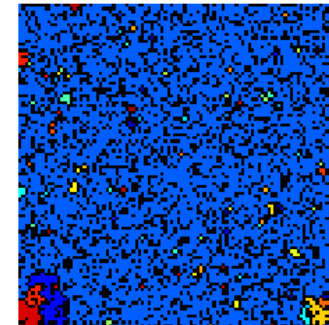
$p=0.65$



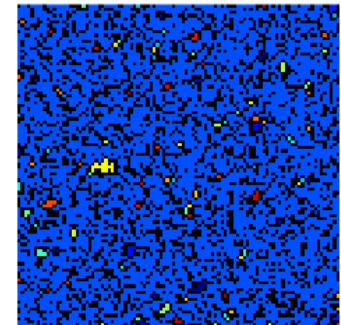
$p=0.65$



$p=0.68$



$p=0.68$

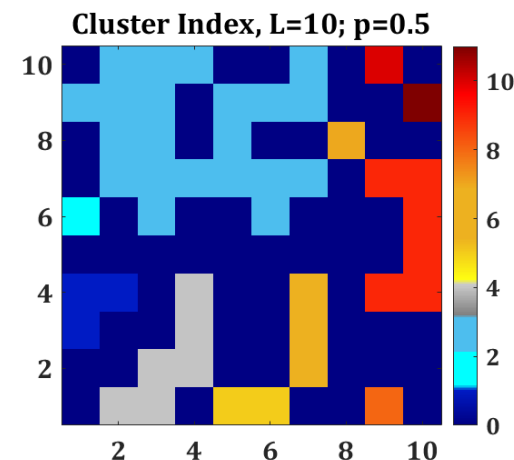
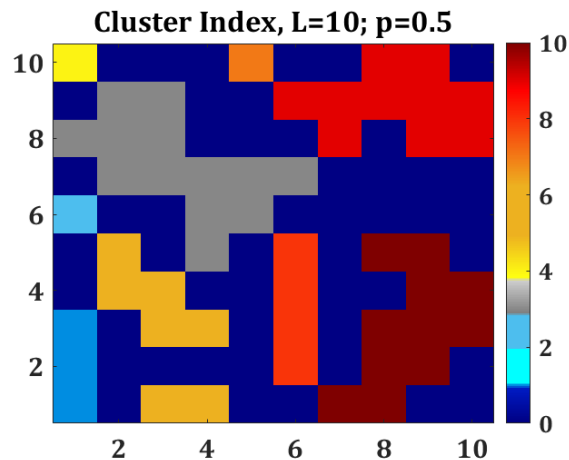


Percolation in 2-d

Q: How many of the sites belong to the biggest cluster?

$P(p)$ is the fraction of sites that belong to the biggest cluster. $P(p)$ is the order parameter

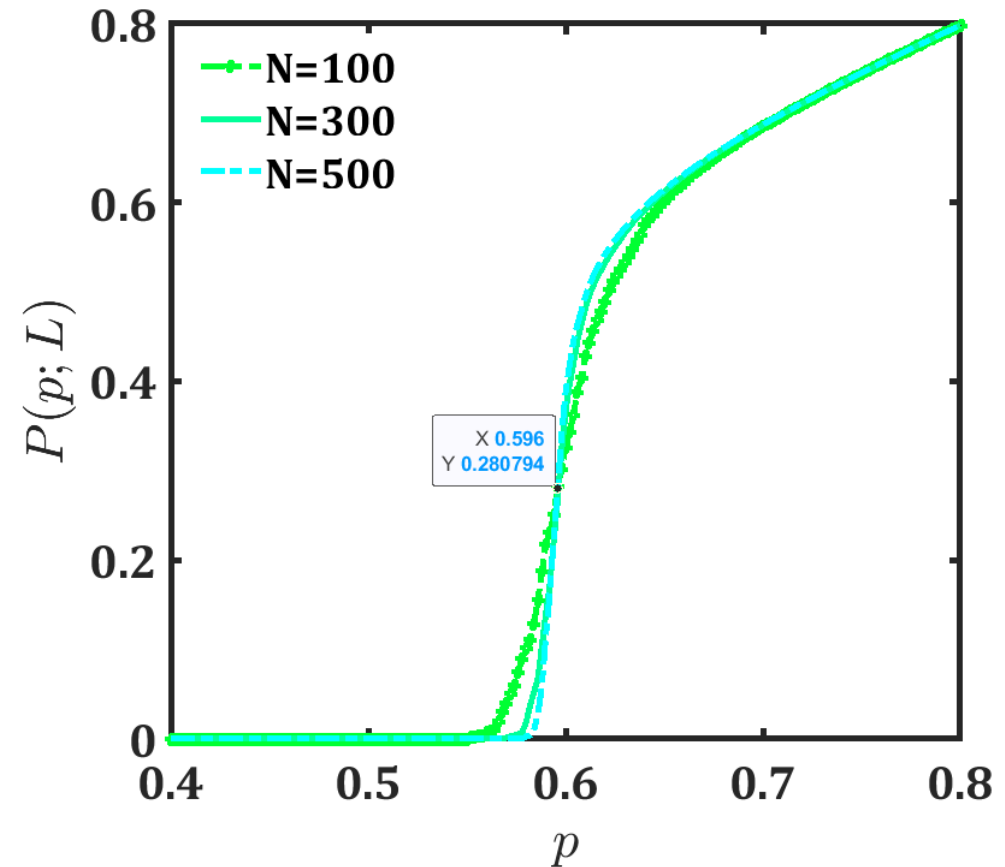
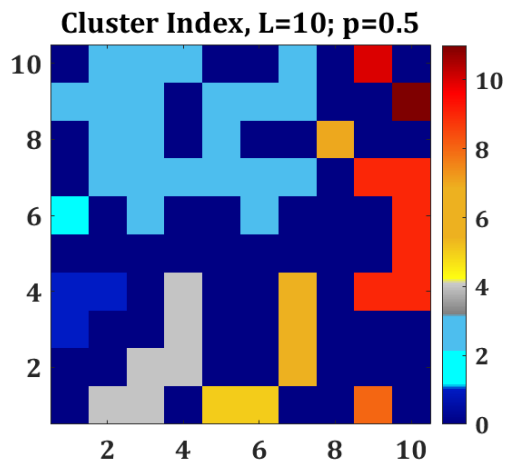
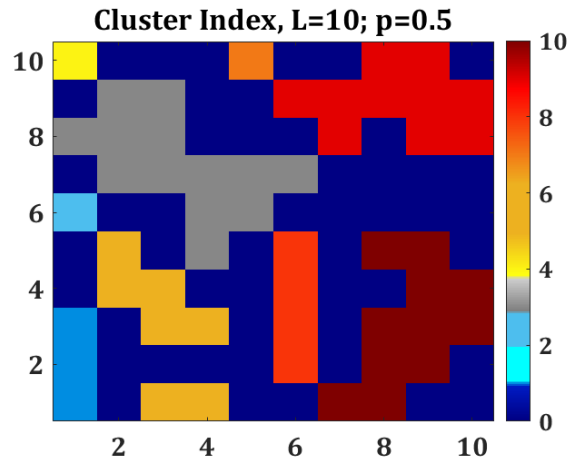
- By creating several random matrices, we can measure $P(p, L)$ in a $L \times L$ lattice.
- For each p , and each realization we identify all the clusters and select the one with the greatest extent.
- Record the number of sites (M) participating in the spanning cluster (Focus in the largest cluster).
- Calculate $\frac{M}{L^2}$ for all p s and for large number of realization.



Percolation in 2-d

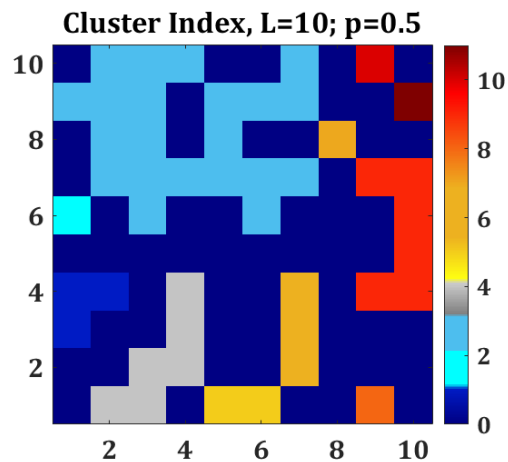
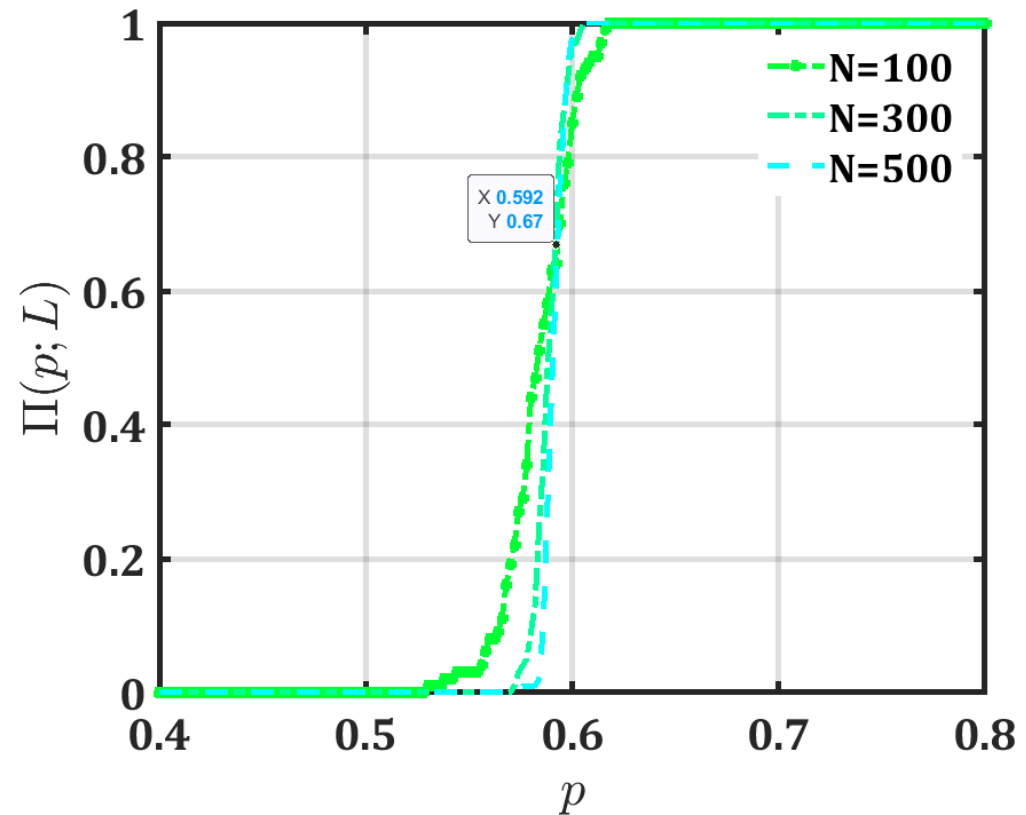
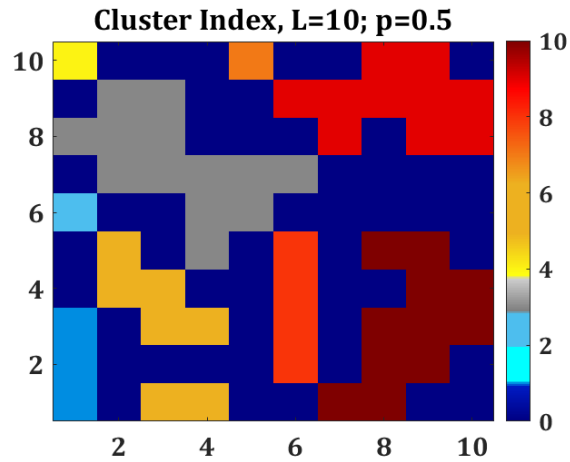
Q: How many of the sites belong to the biggest cluster?

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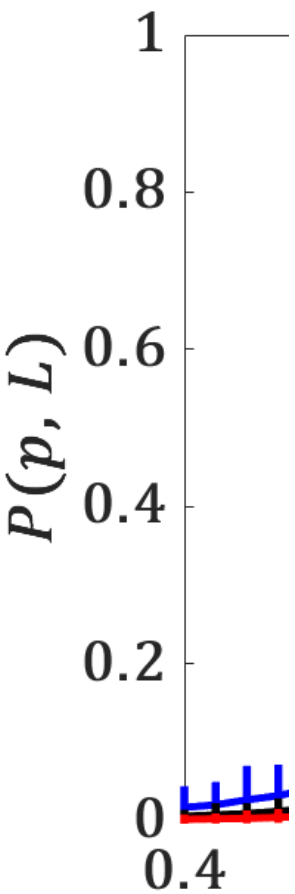
Percolation in 2-d

The **percolation probability** $\Pi(p, L)$ is the probability for there to be a connected path from one side to another side as a function of p in a system of size L .



Percolation in 2-d : Density of the spanning cluster

The probability of the existence of a spanning cluster, $P(p, L)$, as a function of the occupation probability p for various lattices.



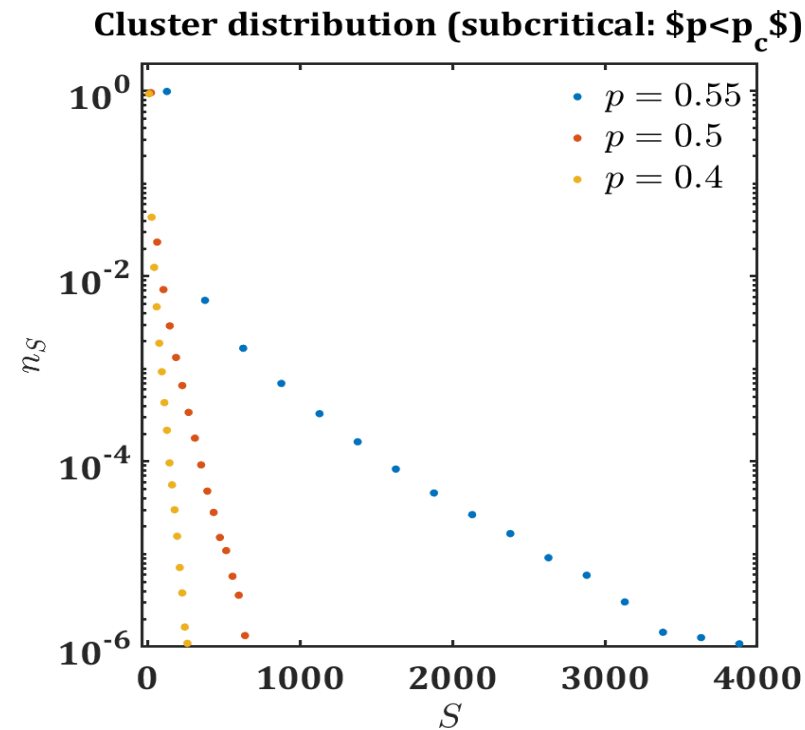
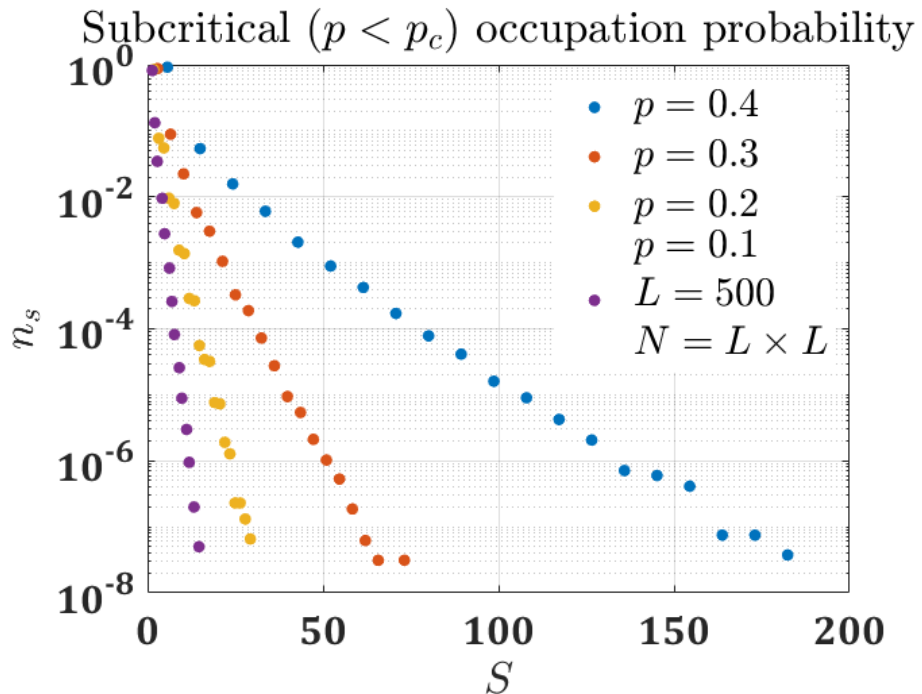
Lattice	Site	Bond
Honeycomb	0.6962	0.65271
Square	0.592746	0.50000
Triangular	0.500000	0.34729
Diamond	0.43	0.388
Simple cubic	0.3116	0.2488
BCC	0.246	0.1803
FCC	0.198	0.119
d = 4 hypercubic	0.197	0.1601
d = 5 hypercubic	0.141	0.1182
d = 6 hypercubic	0.107	0.0942
d = 7 hypercubic	0.089	0.0787

```
Z(indL,5),'b');  
,3),'b', 'linewidth',3);  
5),'k');hold on;  
; ,3),'k', 'linewidth',3);  
ZZ(indL,5),'r');hold on;  
; ,3),'r', 'linewidth',3);
```

end
end

Percolation in 2-d : Cluster size distribution

- Relative cluster size n_s depends on p .
 - For a given p , we take a lattice and repeat the experiments for large realizations ($\sim 10^3$).
 - Lattice is fixed at $N = 500 \times 500$.
 - For a given p , and for a given realization we record all the cluster sizes.
- Q: How the distribution varies from subcritical to supercritical regime (via critical) regime? How does the distribution look like?

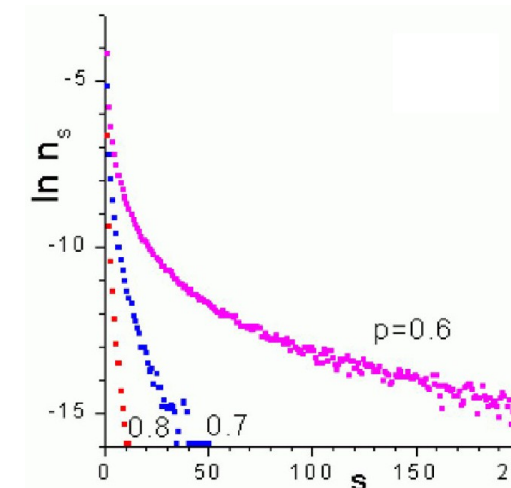
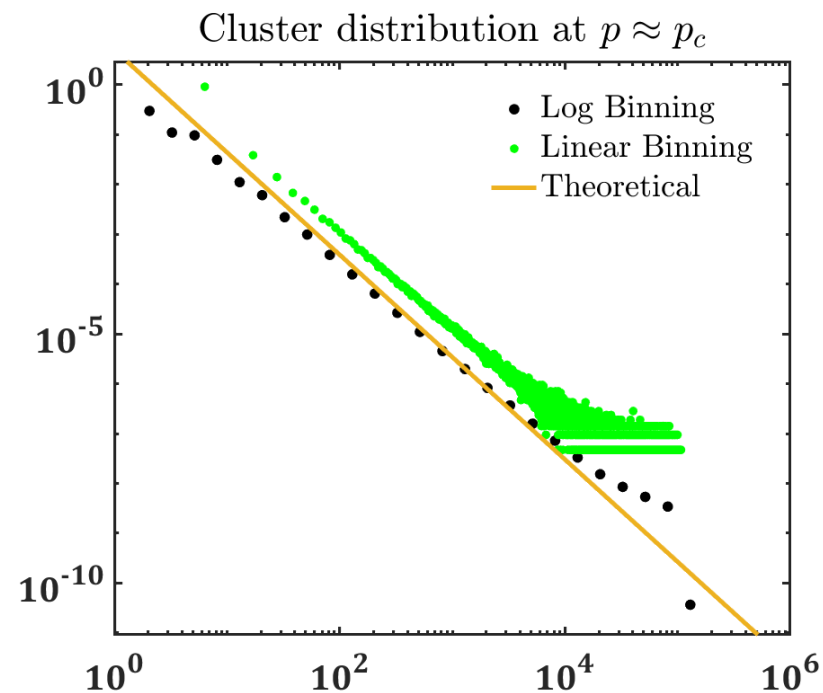
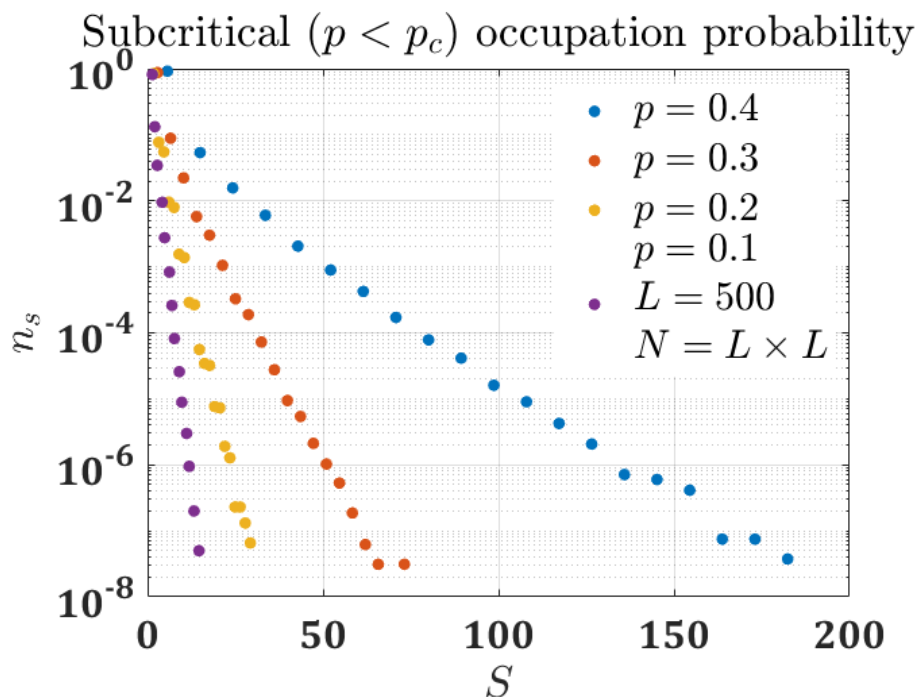


H.J. Herrmann

Anders Malthe-Sorensen

Percolation in 2-d : Cluster size distribution

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H.J. Herrmann

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Percolation in 2-d : Cluster size distribution

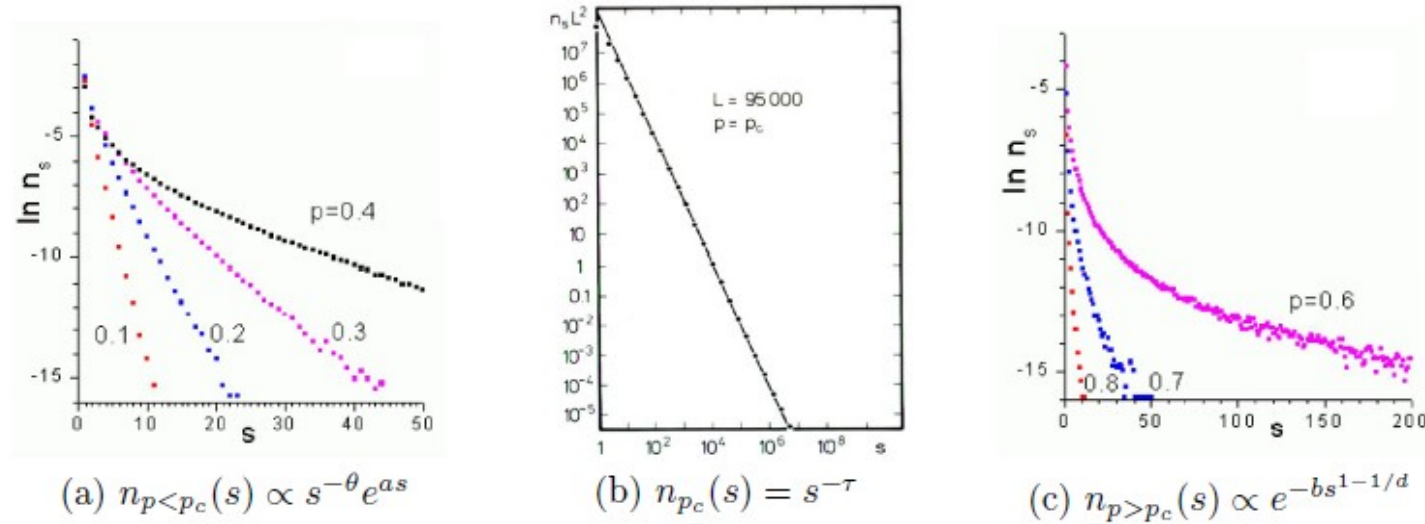


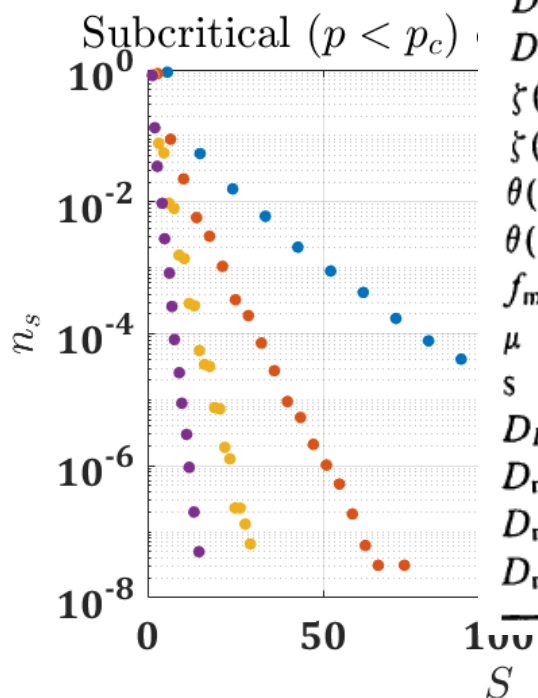
Figure 2.10: Cluster size distribution for $p < p_c$ (left), $p = p_c$ (center) and $p > p_c$ (right). [10]

$$n_p(s) \propto \begin{cases} s^{-\theta} e^{as} & p < p_c \\ s^{-\tau} & p = p_c \\ e^{-bs^{1-1/d}} & p > p_c \end{cases}$$

Percolation in 2-d : Cluster size distribution

- Relative cluster size distribution
- For a given p , the distribution is universal
- Lattice is fixed
- For a given p , the distribution is universal

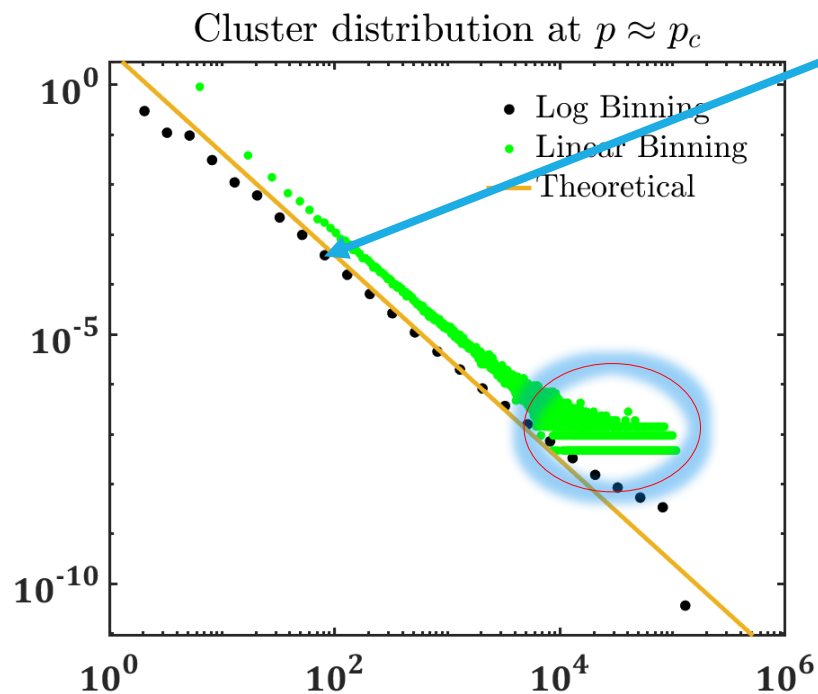
- Q: How the distribution changes with p ?



Exponent	$d = 2$	$d = 3$	$d = 4$	$d = 5$	$d = 6 - \epsilon$	Bethe	Page
α	$-2/3$	-0.62	-0.72	-0.86	$-1 + \epsilon/7$	-1	39
β	$5/36$	0.41	0.64	0.84	$1 - \epsilon/7$	1	37
γ	$43/18$	1.80	1.44	1.18	$1 + \epsilon/7$	1	37
ν	$4/3$	0.88	0.68	0.57	$\frac{1}{2} + 5\epsilon/84$	$1/2$	60
σ	$36/91$	0.45	0.48	0.49	$\frac{1}{2} + O(\epsilon^2)$	$1/2$	35
τ	$187/91$	2.18	2.31	2.41	$\frac{5}{2} - 3\epsilon/14$	$5/2$	33
$D(p = p_c)$	$91/48$	2.53	3.06	3.54	$4 - 10\epsilon/21$	4	10
$D(p < p_c)$	1.56	2	$12/5$	2.8	$-$	4	62
$D(p > p_c)$	2	3	4	5	$-$	4	62
$\zeta(p < p_c)$	1	1	1	1	$-$	1	56
$\zeta(p > p_c)$	$1/2$	$2/3$	$3/4$	$4/5$	$-$	1	56
$\theta(p < p_c)$	1	$3/2$	1.9	2.2	$-$	$5/2$	54
$\theta(p > p_c)$	$5/4$	$-1/9$	$1/8$	$-449/450$	$-$	$5/2$	54
$f_{\max}$	5.0	1.6	1.4	1.1	$-$	1	42
μ	1.30	2.0	2.4	2.7	$3 - 5\epsilon/21$	3	91
s	1.30	0.73	0.4	0.15	$-$	0	93
D_B	1.6	1.74	1.9	2.0	$2 + \epsilon/21$	2	95
$D_{\min}(p = p_c)$	1.13	1.34	1.5	1.8	$2 - \epsilon/6$	2	97
$D_{\min}(p < p_c)$	1.17	1.36	1.5	$-$	$-$	2	98
$D_{\max}(p = p_c)$	1.4	1.6	1.7	1.9	$2 - \epsilon/42$	2	97

e? How does

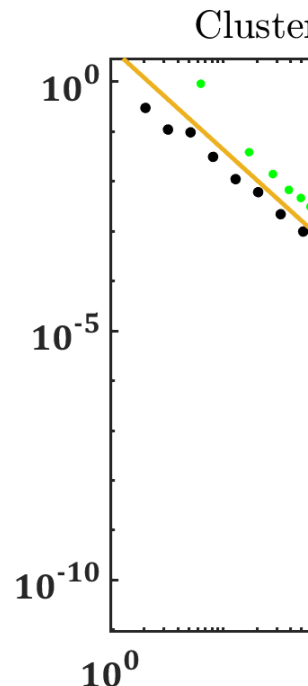
Percolation in 2-d : Cluster size distribution



```
%%^^^log binning^^^%%  
bins = 10.^(0.001:0.2:6);  
[Lob, Lob2] = histcounts(zz(:,1),bins);  
%%^[Loblin, Lob2lin] = hist(zz(:,1),10000);  
ds = diff(bins);  
sl = (bins(1:end-1)+bins(2:end))*0.5;  
nsl = Lob(1:end)./(sum(Lob)*ds);  
loglog(sl,nsl,'ok');  
hold on;
```

we cannot choose a very small bin size for the histogram. However, we see that for small values of s we want to have a small bin size, since the statistics here is good, but for large values of s we want to have larger bin sizes. This is often solved by using logarithmic binning.

Percolation in 2-d : Cluster size distribution



This is often achieved by using logarithmic binning. We choose a basis a^i , where a is the basis for the bins and i is bin number. If we chose $a = 2$ as the basis for the bins, the bin edges will be $2^0, 2^1, 2^2, 2^3, \dots$, that is $1, 2, 4, 8, \dots$ (Maybe we should instead have called the method *exponential binning*). We then count how many events occur in each such bin. If we number the bins by i , then the edges of the bins are $s_i = a^i$, and the width of bin i is $\Delta s_i = s_{i+1} - s_i$. We then count how many event, N_i , occurring in the range from s_i to $s_i + \Delta s_i$, and we use this to find the cluster number density $n(s, p; L)$. However, since we now look at ranges of s values, we need to be precise: We want to measure the probability for a cluster to belong to a specific site of a cluster in the range from s to $s + \Delta s$, that is, we want to measure $n(s, p; L) \Delta s$, which we estimate from

$$\overline{n(s_i, p; L)} \Delta s_i = \frac{N_i}{M L^d}, \quad (4.4)$$

and we can then write

$$\overline{n(s_i, p; L)} = \frac{N_i}{M L^d \Delta s_i}. \quad (4.5)$$

It is important to remember to divide by Δs_i when the bin sizes are not all the same! We implement this by generating an array of all the bin

1. have a small bin
 2. for a given range of s we want
 3. using logarithmic

Percolation in 2-d

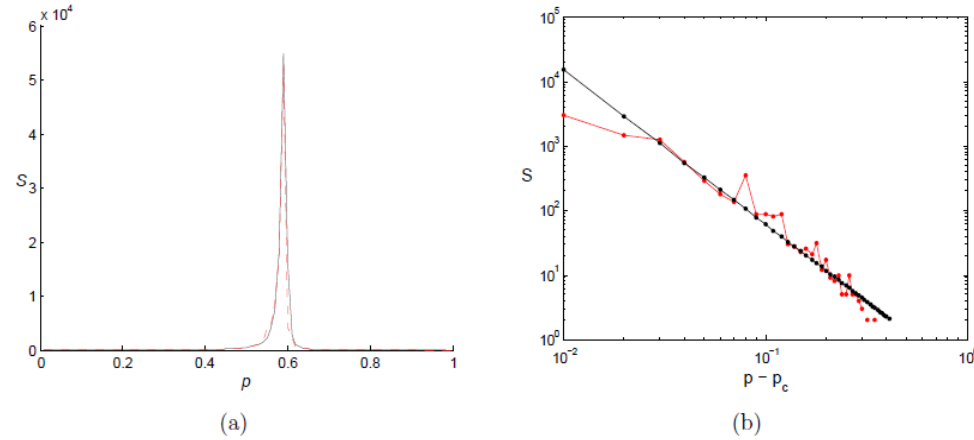


Figure 4.5: Linear plot (a) and loglog plot (b) of the average cluster size S vs p . Figure (a) shows S vs p calculated for one iteration (red dashed line) and averaged data over 1,000 iterations (black solid line) and (b) displays the loglog plot of S vs p with the 1,000 iteration averaged average cluster size data (red line) and the critical exponent line $|p - p_c|^{-\frac{43}{18}}$ (black line) to show the similarity.

2.2.5 Size Dependence of the Order Parameter

We are now going to consider the situation at the critical occupation probability p_c . The size of the largest cluster shall be denoted by s_∞ and the side length of the square lattice by L . When we plot s_∞ against L , we notice that there is a power law at work,

$$s_\infty \propto L^{d_f}$$

where the exponent d_f depends on the dimension of the problem. For a square lattice (2D) we find $d_f = \frac{91}{48}$ and for a three dimensional cube we find $d_f \approx 2.51$. We shall show later on that

$$d_f = d - \frac{\beta}{\nu}$$

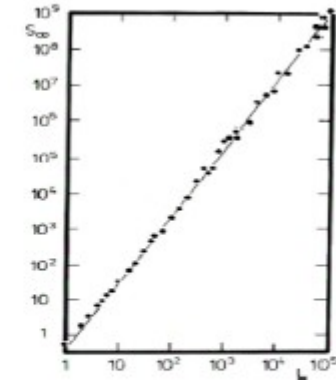


Figure 2.13: Size dependence of the order parameter [10]

<https://www.youtube.com/watch?v=hjGFp7IMi9A>

Percolation in Networks

Random graph: Preliminaries

(Binomial distribution and Poisson distribution)

The probability of getting exactly k successes in n trials is given by the probability mass function:

$$P(k) = \binom{n}{k} p^k (1-p)^{n-k} \text{ where as } \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$$k = 0, 1, 2, \dots, n$$

$$(1-p)^{n-k}$$

$$= 1 - \binom{n-k}{1} p + \binom{n-k}{2} p^2 - \binom{n-k}{3} p^3 + \dots$$

$$= 1 - (n-k)p + \frac{(n-k)(n-k-1)}{2!} p^2 - \frac{(n-k)(n-k-1)(n-k-2)}{3!} p^3 + \dots$$

$$= 1 - np + \frac{(np)^2}{2} - \frac{(np)^3}{3!} + \dots$$

$$= e^{-np}$$

when the number of trials n increases indefinitely whilst the product $\mu = np$, which is the expected value of the number of successes from the trials, remains constant.

$$P(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$= \frac{n!}{k!(n-k)!} p^k e^{-np} = \frac{n(n-1)(n-2)\dots(n-k)!}{k!(n-k)!} e^{-np} p^k = \frac{n^k}{k!} e^{-np} p^k$$

known as the Poisson distribution

Random graph: Preliminaries

(Binomial distribution and Poisson distribution)

Mean/expectation value

$$\begin{aligned}
 E(k) &= \langle k \rangle = \sum_{k=0}^n k f(k) \\
 &= \sum_{k=0}^n \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} k \\
 &= \sum_{k=1}^n \frac{n!}{(k-1)!(n-k)!} p^k (1-p)^{n-k} \\
 &= np \sum_{k=1}^n \frac{(n-1)!}{(k-1)!(n-k)!} p^{k-1} (1-p)^{n-k} \\
 &= np \sum_{k=1}^n \binom{n-1}{k-1} p^{k-1} (1-p)^{n-k} \\
 &= np(1 + \binom{n-1}{1} p(1-p)^{n-1} + \binom{n-1}{2} p^2(1-p)^{n-2} + \dots) \\
 &= np[p + (1-p)]^{n-1} \\
 &= np
 \end{aligned}$$

$$f(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Variance

$$Var(k) = E(k^2) - (E(k))^2 = \langle k^2 \rangle - (\langle k \rangle)^2$$

$$\begin{aligned}
 \langle k^2 \rangle &= \sum_{k=0}^n k^2 f(k) \\
 &= \sum_{k=0}^n \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} k \\
 &= \sum_{k=1}^n \frac{n!k}{(k-1)!(n-k)!} p^k (1-p)^{n-k} ; (k-1=j) \\
 &= \sum_{j=0}^{n-1} \frac{(n)!(j+1)}{j!(n-j-1)!} p^{j+1} (1-p)^{n-(j+1)} \\
 &= \sum_{j=0}^{n-1} \frac{(n)!(j)}{j!(n-j-1)!} p^{j+1} (1-p)^{n-(j+1)} + \sum_{j=0}^{n-1} \frac{(n)!}{j!(n-j-1)!} p^{j+1} (1-p)^{n-(j+1)} \\
 &= p \sum_{j=1}^{n-1} \frac{(n)!}{(j-1)!(n-j-1)!} p^j (1-p)^{n-(j+1)} + np \sum_{j=0}^{n-1} \frac{(n-1)!}{j!(n-1-j)!} p^j (1-p)^{n-1-j} ; (j-1=jj) \\
 &= pn(n-1) \sum_{jj=0}^{n-2} \frac{(n-2)!}{(jj)!(n-2-jj)!} p^{jj+1} (1-p)^{n-2-jj} + np \\
 &= p^2 n(n-1) \sum_{jj=0}^{n-2} \frac{(n-2)!}{(jj)!(n-2-jj)!} p^{jj} (1-p)^{n-2-jj} + np \\
 &= np + n(n-1)p^2 = np + n^2p^2 - np^2 \Rightarrow Var(k) = np - n^2p^2 - np^2 + n^2p^2 = np(1-p)
 \end{aligned}$$

Random graph: Preliminaries

(Binomial distribution and Poisson distribution)

Binomial distribution

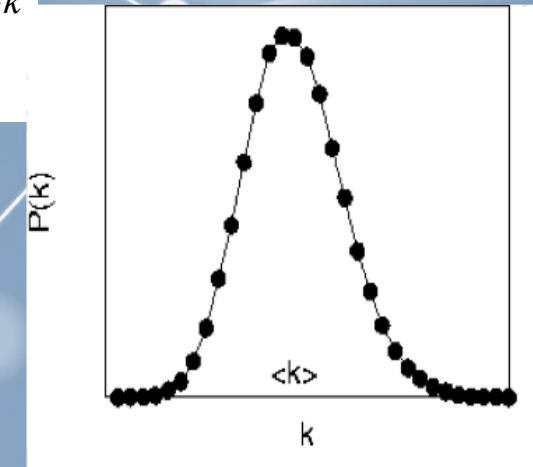
$$P(k) = \binom{N}{k} p^k (1-p)^{N-k}$$

Expectation/Mean

$$\langle k \rangle = Np$$

Variance

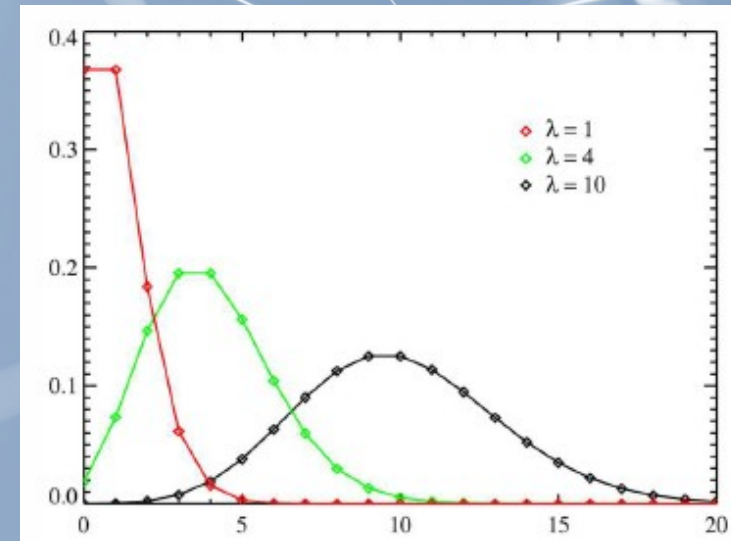
$$\langle k^2 \rangle - \langle k \rangle^2 = p(1-p)N$$



Poisson distribution

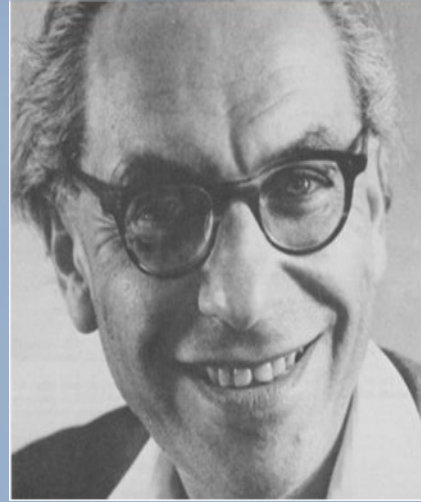
$$P(k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

$$\text{Variance} = \text{Mean} = \lambda$$

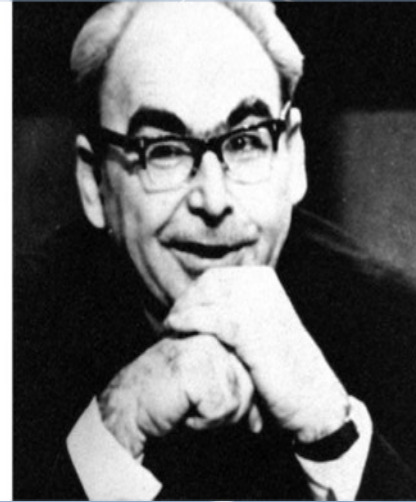


Random Graph: An insight within

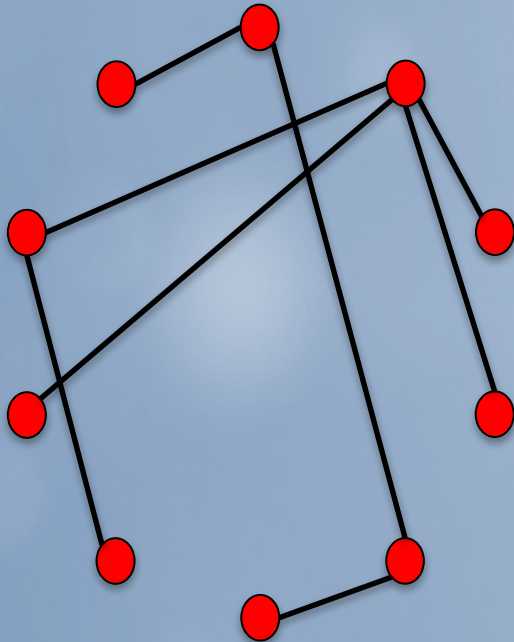
Pál Erdős
(1913-1996)



Alfréd Rényi
(1921-1970)



Construction



Erdős-Rényi model (1960)

Connect with probability p

$$p = 1/6 \quad N = 10$$
$$\langle k \rangle \sim 1.5$$

Total number of connections =

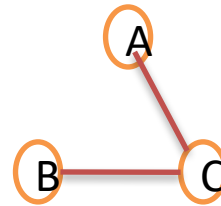
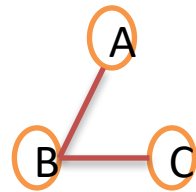
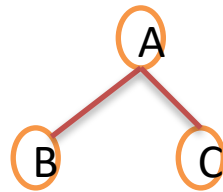
$$\frac{10 \times 9}{2} = 45$$

Possible number of connections with probability $1/6$
 $= 45 \times 1/6 = 8$

Random Graph: An insight within

- In the $G(n, M)$ model, a graph is chosen uniformly at random from the collection of all graphs which have n nodes and M edges.*

Construction



- In the $G(n, p)$ model, a graph is constructed by connecting nodes randomly. Each edge is included in the graph with probability p independent from every other edge.*

Erdos–Renyi random graphs (ER)

- *Model a network $G(V,E)$ with $|V|=n$ and $|E|=m$*
- *An ER graph that models G is constructed as follows:*
 - *It has n nodes*
 - *Edges are added between pairs of nodes uniformly at random with the same probability p*
 - *Two (equivalent) methods for constructing ER graphs:*
 - *$G_{N,p}$: pick p so that the resulting model network has m edges*
 - *$G_{N,m}$: pick randomly m pairs of nodes and add edges between them with probability 1*

Properties of Random Network

As the network size increases, the distribution becomes increasingly narrow—we are increasingly confident that the degree of a node is in the vicinity of $\langle k \rangle$.

$$P(k) = \binom{N-1}{k} p^k (1-p)^{(N-1)-k}$$

Select k
nodes from $N-1$

probability of
having k edges

probability of missing
 $N-1-k$ edges

$$\langle k \rangle = p(N-1)$$

$$\sigma_k^2 = p(1-p)(N-1)$$

$$\frac{\sigma_k}{\langle k \rangle} = \left[\frac{1-p}{p} \frac{1}{(N-1)} \right]^{1/2} \approx \frac{1}{(N-1)^{1/2}}$$

As the network size increases, the distribution becomes increasingly narrow—we are increasingly confident that the degree of a node is in the vicinity of $\langle k \rangle$.

**Percolation process: emergence
of giant component**

Properties of Random Network

- If $pN < 1$ ($Np = c = \langle k \rangle$), then a graph in $G(N, p)$ will almost surely have no connected components of size larger than $O(\log(N))$.
- If $pN = 1$, then a graph in $G(N, p)$ will almost surely have a largest component whose size is of order $N^{2/3}$.
- If $pN \rightarrow c > 1$, where c is a constant, then a graph in $G(N, p)$ will almost surely have a unique giant component containing a positive fraction of the vertices. No other component will contain more than $O(\log(N))$ vertices.

Properties of Random Network

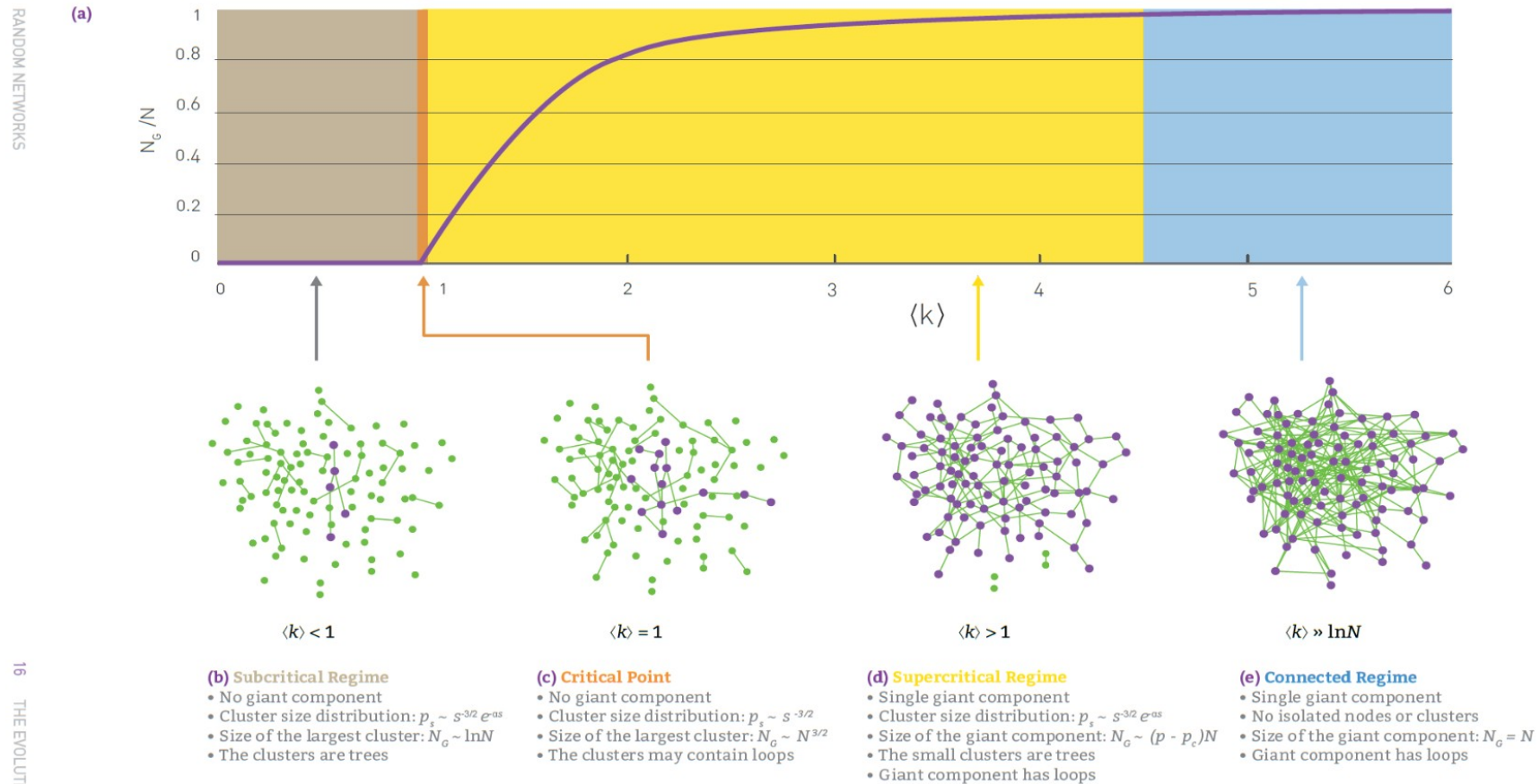


Figure 3.7
Evolution of a Random Network

(a) The relative size of the giant component in function of the average degree $\langle k \rangle$ in the Erdős-Rényi model. The figure illustrates the phase transition at $\langle k \rangle = 1$, responsible for the emergence of a giant component with nonzero N_g .

(b-e) A sample network and its properties in the four regimes that characterize a random network.

Giant Component

- A network component whose size grows in proportion to n is called giant component
- Let u be the fraction of nodes that do not belong to the giant component. Hence,
 - If there is no giant component $\rightarrow u = 1$
 - If there is giant component $\rightarrow u < 1$
- In order for a node i not to connect to the giant component:
 - i needs not connect to any other node $j \rightarrow 1 - p$
 - i is connected to j , but j itself is not connected to the giant component $\rightarrow pu$

Giant Component

- Hence, the total probability of i not being connected to giant component via vertex j is: $1 - p + pu$
 - Considering all $n - 1$ vertices through which i can connect:

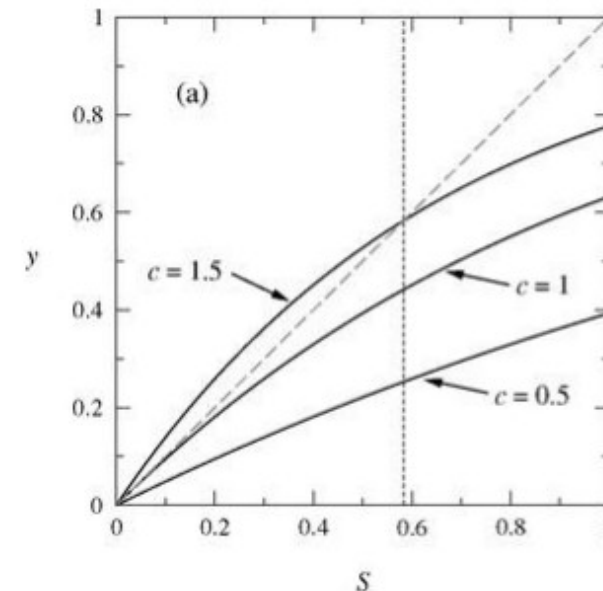
$$u = (1 - p + up)^{n-1} = \left[1 - \frac{c}{n-1}(1-u) \right]^{n-1} \quad c \text{ is the average degree}$$

- Taking the logarithms at both sides and the Taylor approximation for large n :
- This equation cannot be solved in close form

$$u = e^{-c(1-u)} \xrightarrow{S=1-u} S = 1 - e^{-cS}$$

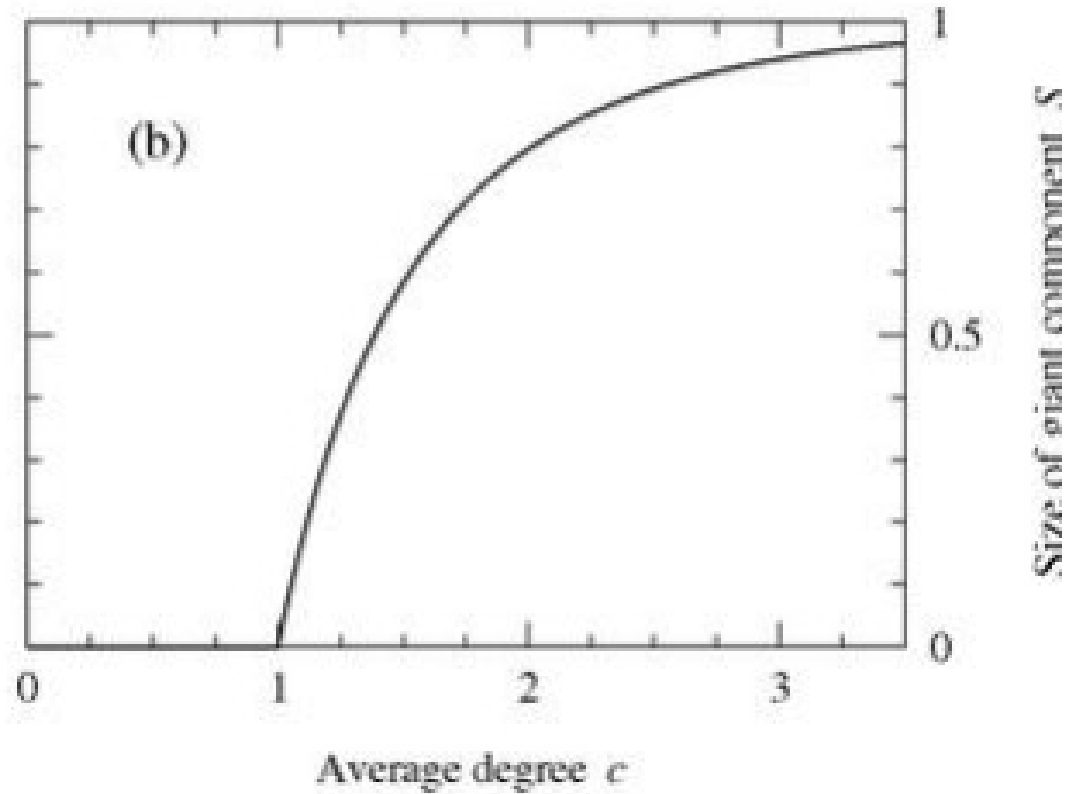
Giant Component

- We plot $y = 1 - e^{-cS}$ with S between 0 and 1 (since it represents fraction of nodes)
- We also plot $y = S$
- The point where the two curves intersect is the solution
- For small c only one solution
 - $S = 0$
- For greater c there might be two solutions
 - The point where two solutions start appearing is when the gradients of the curves are equal at $S = 0$
 - ✓ This happens for $c = 1$



Giant Component

- We expect a giant component iff $c > 1$



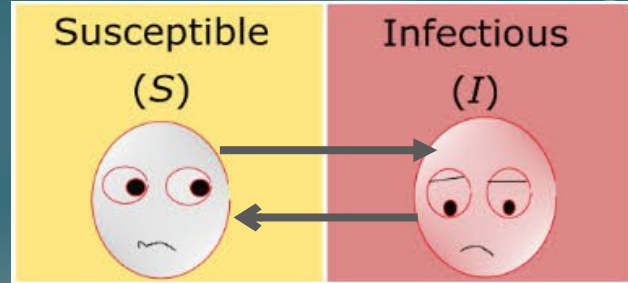
Epidemic Spreading : Modelling

SIS Dynamics

$$\frac{ds}{dt} = \gamma i - \beta s i$$

$$\frac{di}{dt} = \beta s i - \gamma i$$

$$s + i = 1 \text{ (Constant)}$$



$$\frac{1}{p} \left(\log \left(\frac{p - qx}{x} \right) - \log \left(\frac{p - qx_0}{x_0} \right) \right) = -t$$

$$p = (\beta - \gamma); q = \beta$$

$$\left(\frac{p - qx}{x} \right) = e^{-pt} \left(\frac{p - qx_0}{x_0} \right)$$

$$\frac{p}{x} = q + e^{-pt} \left(\frac{p - qx_0}{x_0} \right)$$

$$x = \frac{p}{q + e^{-pt} \left(\frac{p - qx_0}{x_0} \right)}$$

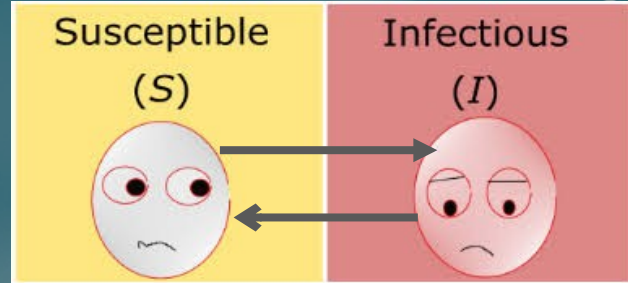
Epidemic Spreading : Modelling

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$$\frac{di}{dt} = \beta s i - \gamma i$$

$$s + i = 1 \text{ (Constant)}$$



$$p = (\beta - \gamma); q = \beta$$

$$i = \frac{\frac{\beta - \gamma}{\beta}}{1 + \left(\frac{\beta - \gamma}{\beta i_0} - 1 \right) e^{-(\beta - \gamma)t}}$$

$$i_{ee} = \frac{\beta - \gamma}{\beta} \text{ endemic equilibrium, } t \rightarrow \infty$$